



NEWS 73 *December 2013*

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FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

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Contents

Editorial
 Announcements
 Research in progress
 New books
 Featured research
 Fleas in art and history
 Mark Twain on fleas
 Gerard ter Borch
 Flea research of historical interest
 Directory of Siphonapterists
 Siphonaptera Literature 2013
 Obituaries
 Robert “Bob” Stephen George (1921-2013), UK Flea Recorder
 Publications of K. Hůrka (1931-2004) involving bat parasites
 Nixon Wilson (1930-2011), Ectoparasitologist

Editorial

Dear Flea News Reader,

It is with sadness to place obituaries of Robert George, Karel Hůrka, and Nixon Wilson in this Flea News.

The last Flea News (Volume 72, June 2013) featured several reports on giant fossil fleas (Gao et al. 2012, Huang et al. 2012, 2013; see featured research in this Flea News for their abstracts). If these earliest fossils are really of fleas, then fleas seem to have originated well before mammals and birds, and presumably parasitized dinosaurs, or perhaps reptilian synapsids. The synapsids are akin to mammals, sharing similar skull foramina.

However, paleontologists have found flea-like fossils before that turned out not to be true fleas. It seems likely that dinosaurs had their own ectoparasites, but we find diverse insects are ectoparasitic, including the lice, bugs, and flies such as the sheep keds and bat flies. One major difference between these putative flea fossils and Siphonaptera is that the fossils are dorsoventrally compressed and do not appear to have jumped, contrasting with the Siphonapteran lateral compression and jumping.

A new, living snow scorpionfly (Boreidae) discovered in Alaska (see featured research, Sikes and Stockbridge 2013) shows many flea-like features, including the ability to jump. Hypothetically, true fleas may have originated in vertebrate nests in cold,

boreal forests. This does not rule out the dinosaurs though, because surprisingly dinosaurs had elaborate nests and seem to have traipsed and gadded about through these temperate areas as well. Further studies on these interesting insects from the past and present are required to determine how they relate to Siphonaptera.

Changing the subject again, I have noted a societal deterioration in information access and copyright protection. Increasingly, at least in the United States, public universities are restricting and monitoring the public's access to taxpayer-supported databases. Additionally, in publishing, "creative commons" journals ask authors to donate their copyright to public domain, or corporate-published journals demand authors transfer their copyright to the corporation. While these may seem like opposing publishing models, both are the same in removing from the individual copyright protection and possible fruits of his or her creation.

Access to publicly funded databases, and copyright protection available for the lifetime of individual creators, are simply essential to thriving academic freedom and creative work. These issues should be of interest to any creative citizen in a democracy.

I received a few items that are not in this Flea News, as they are in Flea News previously. Thanks to everyone who sent me items, and helped with editing! Happy New Year 2014!

Best regards,
R.L. Bossard
Editor, Flea News

Announcements

New Doctoral Dissertations and Masters' Theses

Student: MEGAN M. FRIGGENS

Title: FLEAS, HOSTS AND HABITAT: WHAT CAN WE PREDICT ABOUT THE SPREAD OF VECTOR-BORNE ZOONOTIC DISEASES?

Dissertation, Doctor of Philosophy, School of Forestry, Northern Arizona University, USA, May 2010.

Vector-borne diseases of humans and wildlife are experiencing resurgence across the globe. I examine the dynamics of flea-borne diseases through a comparative analysis of flea literature and analyses of field data collected from three sites in New Mexico: The Sevilleta National Wildlife Refuge, the Sandia Mountains, and the Valles Caldera National Preserve (VCNP). My objectives were to use these analyses to better predict and

manage for the spread of diseases such as plague (*Yersinia pestis*).

To assess the impact of anthropogenic disturbance on flea communities, I compiled and analyzed data from 63 published empirical studies. Anthropogenic disturbance is associated with conditions conducive to increased transmission of flea-borne diseases. Most measures of flea infestation increased with increasing disturbance or peaked at intermediate levels of disturbance. Future trends of habitat and climate change will probably favor the spread of flea-borne disease.

Rodents, including Gunnison's prairie dogs (*Cynomys gunnisoni*), were trapped for three years (2004-2006). Blood and flea samples were tested for the presence of plague and another bacterial pathogen, *Bartonella*. I conduct two analyses with this data. The first examines prairie dogs and their flea communities in the VCNP. Prairie dogs experienced a plague epizootic in fall 2004, after which we found plague positive fleas and positive antibody titers in three prairie dogs. We noted an increased tendency for flea exchange opportunities in the spring before flea abundance peaked. Spring conditions, which favor presence and exchange of certain flea species, may be just as important for determining plague outbreaks as the summer conditions, which lead to build up in flea populations.

In the second analyses, I found 38% of the rodents of 30 species and 60% of fleas of 24 species positive for *Bartonella*. *Bartonella* infections typically lasted two months and the prevalence of *Bartonella*. [Editor's note: see author's comment below.] Changes in prevalence related to host density and environmental gradients, point to the importance of both fleas and rodents in *Bartonella* transmission cycles.

This research shows environment influences the risk of flea-borne disease spread. It is likely that future trends of habitat and climate change will favor the spread of flea-borne diseases, including plague and *Bartonella*.

[Author's comment, received 14 January 2014: *Bartonella* infections typically lasted two months, though three animals tested positive for three consecutive months. The prevalence of *Bartonella* corresponded with rodent density at each site, though the nature of this relationship changed with season and elevation. Analysis of temporal patterns of *Bartonella* infection found no significant effect of sampling period, though sites were significantly different.]

Friggens, M. M., & Beier, P. (2010). Anthropogenic disturbance and the risk of flea-borne disease transmission. *Oecologia*, 164 (3), 809-820.

Friggens, M. M., Parmenter, R. R., Ford, P. L., Boyden, M., Gage, K., and P. Kiem. 2010. Flea abundance, diversity, and plague in Gunnison's prairie dogs (*Cynomys gunnisoni*) and their burrows in montane grasslands in northern New Mexico. *Journal of Wildlife Disease* 4(2): 356-367.

Friggens, MM; Ford, PL; Parmenter, RR; Boyden, M.; Gage, K. 2011. Assessing plague risk and presence through surveys of small mammal flea communities. In: Feldman, Susana R.; Oliva, Gabriel E.; Sacido, Monica B., eds. IX International Rangeland

Congress: Diverse Rangelands for a Sustainable Society; April 3-10, 2011; Rosario, Argentina. IRC2001 Congress.

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Student: Katy L. Parise

Title: High *Yersinia pestis* loads significantly reduce bacterial diversity in wild-caught fleas

M.S. Thesis, Biological Sciences, Northern Arizona University, USA, 2012.

The bacterial community associated with fleas may influence the transmission of specific pathogens, including *Yersinia pestis*, the etiological agent of plague. Although the flea microbiome previously has been surveyed to characterize bacterial relationships with specific endosymbionts and with *Y. pestis*-infected fleas under laboratory conditions, it has not been studied in wild-caught, naturally *Y. pestis*-infected fleas. To characterize the microbiome under these conditions, bacterial communities were surveyed in a population of *Oropsylla hirsuta* collected from Gunnison's prairie dog burrows during a plague epizootic. Barcoded DNA pyrosequencing was used to determine the bacterial abundance, richness, and diversity of *Y. pestis*-positive and -negative fleas. *Y. pestis* loads were quantified to better understand the level at which *Y. pestis* might alter the community.

Bacterial communities with high loads of *Y. pestis* were significantly more similar to one another than low-infected and *Y. pestis*-negative communities ($R = 0.59$; $p = 0.001$), suggesting that the presence of high *Y. pestis* loads significantly reduces the abundance of or completely eliminates other flea-associated bacterial lineages. Bacterial communities across different sites, times, and flea species also were compared, and a core community of bacterial lineages was determined to reside in all fleas. Additionally, to evaluate the effects of surface-sterilization on characterizing *Y. pestis*-positive and -negative flea microbiomes, bacterial communities of washed and unwashed fleas within these categories were compared. A significant difference between washed and unwashed *Y. pestis*-negative fleas was determined ($R = 0.63$; $p = 0.009$); however, there was no substantial change between *Y. pestis*-positive fleas (5.01%), suggesting there is an observable effect when surface-sterilization is conducted on an overall species-rich environment. This is the first study to evaluate the effects of surface-sterilization on *Y. pestis*-positive and -negative bacterial communities, as well as present data from wild-caught, naturally infected fleas that suggests *Y. pestis* can dominate the community and thereby procure the greatest opportunity for transmission.

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Student: Ashley M. Winker

Title: PARASITE COMMUNITIES OF TROPICAL FOREST RODENTS:
INFLUENCES OF MICROHABITAT STRUCTURE AND SPECIALIZATION.

Master of Science-Biology Thesis, University of Wisconsin, Oshkosh, USA, May 2013.

Parasitism is the most common life style and has important implications for the ecology and evolution of hosts. Most organisms host multiple species of parasites, and parasite communities are frequently influenced by the degree of host specialization. Parasite communities are also influenced by their habitat – both the host itself and the habitat that the host occupies. Tropical forest rodents are ideal for examining hypotheses relating parasite community composition to host habitat and host specialization. *Proechimys semispinosus* and *Hoplomys gymnurus* are morphologically similar echimyid rodents; however, *P. semispinosus* is more generalized, occupying a wider range of habitats. I predicted that *P. semispinosus* hosts a broader range of parasite species that are less host-specific than does *H. gymnurus* and that parasite communities of *P. semispinosus* are related to microhabitat structure, host density, and season.

During two dry and wet seasons, individuals of the two rodent species were trapped along streams in central Panama to compare their parasites, and *P. semispinosus* was sampled on six plots of varying microhabitat structure in contiguous lowland forest to compare parasite loads to microhabitat structure. Such structure was quantified by measuring thirteen microhabitat variables, and dimensions were reduced to a smaller subset using factor analysis to define overall structure. Ectoparasites were collected from each individual, and blood smears were obtained to screen for filarial worms and trypanosomes.

In support of my prediction, the habitat generalist (*P. semispinosus*) hosted more individual fleas, mites, and microfilaria; contrary to my prediction, the habitat specialist (*H. gymnurus*) hosted more individual lice, ticks, and species of ticks. Also contrary to my prediction, none of the tick species found on *P. semispinosus* were host-specific. The sole flea species I collected was *Polygenis klagesi*, which may be host-specific largely to *P. semispinosus*, only rarely infesting other mammals. Fleas were associated with forest openness with respect to trees and were more abundant during the rainy season. Lice were more abundant during the dry season, and ticks were more abundant on male hosts. Male-biased parasitism is common in mammals and presumably results from greater mobility and lower immune response than females. This descriptive study is the first to investigate the relationships between parasite communities and microhabitat, host density, and season in a lowland Neotropical forest. It lays the foundation for an experimental approach to study the interactions of these hosts and their parasites.

[Editor's note: "All fleas were *Polygenis klagesi*; 50.0% of *P. semispinosus* were infested with fleas, and the mean intensity of infestation was 4.3 fleas/rat." (pg. 10), "In this study, fleas found on *P. semispinosus* were more abundant during the rainy season and were associated with forest openness (i.e., negatively associated with tree density)." (pg. 13)].

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Editor's note: Flea News readers were interested in the thesis abstract of J.

Sanchez in the last Flea News 72 as to what was meant by the *Neotyphloceras* taxonomy and 'species group'. Following is the publication that seems to address this:

JULIANA SANCHEZ, VERÓNICA AMOR, ENRIQUE A. BAZÁN-LEÓN, RODRIGO A. VÁSQUEZ, & MARCELA LARESCHI. (2012) Redescription of *Neotyphloceras chilensis* Jordan, new status (Siphonaptera: Ctenophthalmidae: Neotyphloceratini) *Zootaxa* 3259: 51–57.

Neotyphloceras crassispina chilensis Jordan is elevated to species status based on comparison of its morphology with other species and subspecies of the genus. *Neotyphloceras chilensis* is redescribed from specimens from Region IV of Coquimbo, Chile. A key for identification of species and subspecies of *Neotyphloceras* is provided.

[Extended excerpt from the editor]: "The genus *Neotyphloceras* Rothschild includes two species, *Neotyphloceras rosenbergi* (Rothschild) and *Neotyphloceras crassispina* Rothschild. Three subspecies were recognized in the latter species, distinguishable by differences in the length and shape of the fixed process of the clasper, and the distance of the most distal seta to its apex (Jordan 1936; Hopkins & Rothschild 1966). These subspecies are *Neotyphloceras crassispina crassispina* Rothschild, *N. crassispina hemisus* Jordan, and *N. crassispina chilensis* Jordan. Moreover, Smit (1968) proposed to distinguish females on the basis of the number and location of the upper lateral seta of tergum VIII. However, this characteristic was not considered subsequently, and subspecific identification of females was based on accompanying males (Hastriter 2001; Lareschi *et al.* 2010). According to Hastriter (2001), the distribution of *N. crassispina* subspecies extends from northern Peru to central Bolivia, and south through central Chile and western Argentina, as also noted by others (Smit 1968; Sanchez *et al.* 2009; Lareschi *et al.* 2003, 2010). The modified segments of *Neotyphloceras* species and subspecies are very complex and they have never been studied in sufficient detail to determine the validity of the subspecific taxa (Hastriter 2001), which has been questioned (Beaucournu & Castro 2003). Based on a comparative study of specimens of both sexes of all species and subspecies of *Neotyphloceras*, which includes modified segments, we raise *N. c. chilensis* to species status and redescribe it based on specimens from the IV Region of Coquimbo, Chile."

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Presentations on fleas

Flea presentations at the 2013 Entomological Society of America Annual Meeting

"Allergic responses to fleas, lice, ticks and mites"

"Delusory Parasitosis – When teachers and staff think they have bugs, but they don't"

Nancy Hinkle , University of Georgia, Athens, GA

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"Roles of insects in the mesozoic ecosystems of Northeastern China"

Mei Wang , Capital Normal University, China, Beijing, China

Chungkun Shih , Key Laboratory of Insect Evolution & Environmental Changes, Capital Normal University, Beijing, China

Chen Wang , Capital Normal University, Beijing, China

Dong Ren , Capital Normal University, Beijing, China

Insects have been in existence for at least 400 million years. These relatively small-sized terrestrial arthropods, along with their durable cuticle, have allowed them to be preserved in compression fossils and ambers. Based on insect fossils collected in northeastern China (i.e. Liaoning, Inner Mongolia and Hebei), at least 19 orders and nearly 300 species have been reported (Ren et al. 2010). Paleoentomology in China have made significant progress in recent years.

Many insects have close interactions with plants, which are recorded on fossils. For example, *Protonemestrius jurassicus* had long siphonate mouthpart, suggesting its “flower” visiting habits, which is important for the study of the origin of angiosperms (Ren, 1998). A probable pollination mode before angiosperms was proposed based on the Eurasian, long-proboscid fossil scorpionflies from the Middle Jurassic to the Early Cretaceous, which fed on ovular secretions of extinct gymnosperms and engaged in pollination mutualisms with gymnosperms (Ren et al. 2009). Two extraordinary fossil lacewings demonstrated a preangiosperm origin for leaf mimicry, revealing a lost evolutionary scenario of interactions between insects and gymnosperms (Wang et al., 2010). Lately, a near-perfect mimetic association and potential mutualism between a mecopteran species and a ginkgoalean plant species from the late Middle Jurassic of northeastern China has been discovered, suggesting that hangingflies developed leaf mimicry either as an antipredator avoidance device or possibly as a predatory strategy to provide an antiherbivore function for its plant hosts (Wang et al., 2012). Besides feeding on leaves and plant matters, a katydid, *Archaboilus musicus* was reported to produce low-pitched musical songs, which are well-adapted to communication in the mid-Jurassic forest with coniferous trees and giant ferns (Gu et al., 2012).

On insect association with vertebrates, two well-preserved fossil ectoparasitic insects have been found in the mid-Mesozoic of China, exhibiting many features of ectoparasites. Large body size and long serrated stylets for piercing tough and thick skin or hides of hosts suggest that these primitive ectoparasites might have lived on and sucked the blood of relatively large hosts, such as contemporaneous feathered dinosaurs and/or pterosaurs or medium-sized mammals (Gao et al., 2012). A new transitional flea, *Saurophthyrus exquisitus*, has a series of translational characters in between those of Pseudopulicidae and living fleas suggesting it might have lived and fed on blood of the pterosaurs or other host vertebrates (Gao et al., 2013).

Based on the coexisting plants and other animals, we established a structure of the ecosystems of the Yanliao Biota and Jehol Biota respectively. Insects played important

roles in maintaining food chains and ecological successions, in circulation of substances and global flow of energy.

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Flea presentations at the AMERICAN SOCIETY OF MAMMALOGISTS, 93rd ANNUAL MEETING, Jun 14-18, 2013. Philadelphia, Pennsylvania.

"Large Herbivore Removal Increases Rodent-borne Disease Prevalence in East Africa"

Hillary S. Young 1, Katharina Dittmar de la Cruz 2, Douglas J. McCauley 3, Sarah Billeter 4, Truman P. Young 5, Kristofer M. Helgen 1.

1Smithsonian Institution, Division of Mammals, Washington DC 20013 USA; 2SUNY Buffalo, Department of Biological Sciences, Buffalo NY 14260 USA; 3Department of Environmental Science, Policy, and Management, University of California at Berkeley, Berkeley, CA 94720 USA; 4Center for Disease Control, Fort Collins, CO 80521 USA; 5Department of Plant Sciences, University of California, Davis, CA 95616 USA.

Understanding the effects of wildlife loss on zoonotic disease is important both for public health and conservation initiatives. For rodent-borne diseases, there is strong conceptual support, but little empirical evidence, for the hypothesis that the loss of large wildlife tends to increase rodent-borne disease risk by removing controls on rodent density - either by removing predators or competitors of rodents.

To examine this question, we experimentally excluded large wildlife and then examined the prevalence of *Bartonella* infection in rodents and their ectoparasite vectors and the total density of infected rodents and vectors. We found no significant effect of wildlife removal on prevalence of infection in either rodents or vectors.

However, because rodent and vector abundance roughly doubles following the removal of large wildlife, we found that the density of infected hosts and infected vectors was more than 2 times higher in sites where wildlife was present. As the fleas in this study system also parasitize humans and are known to transmit *Bartonella*, this work strongly suggests that wildlife loss will greatly increase human risk for *Bartonella* transmission.

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"Mitigation Efforts to Conserve Black-footed Ferrets during a Plague Epizootic"

Travis M. Livieri 1, Dean E. Biggins 2, Randall L. Griebel 3, Tonie E. Rocke 4

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Black-footed ferrets (*Mustela nigripes*) are endangered obligate carnivores of prairie dogs (*Cynomys*), and both species are highly susceptible to sylvatic plague. The reintroduced population of ferrets at Conata Basin, South Dakota, was once the largest black-footed ferret population in the world partly due to a lack of plague. In 2008, plague entered Conata Basin and has since affected 20,000+ acres of prairie dogs and ferrets. Dusting prairie dog burrows with an insecticide to kill fleas, a vector of plague, has been successful in maintaining approximately 10,000 acres of habitat. Past studies suggested that dusting alone may be enough to preserve prairie dogs and ferrets in the presence of plague, but we also vaccinated many ferrets against plague. We analyzed data from annual ferret population surveys using logistic regression to compare re-encounter rates of ferrets pre-/post-plague and to assess vaccine effect on dusted colonies. Re-encounter rates were lower in the presence of plague ($P < 0.001$) and males were more susceptible than females ($P = 0.039$). On dusted colonies during a plague epizootic re-encounter rates were higher for vaccinated than nonvaccinated ferrets ($P = 0.078$), with a trend towards even higher survival for vaccinates receiving more than 1 dose of vaccine. The combination of dusting and vaccine is needed for plague mitigation at Conata Basin and without both the population might have perished.

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Meeting announcements

National Conference on Urban Entomology

The 2014 National Conference on Urban Entomology will be held May 18-21, 2014 at the Embassy Suites on the river walk in San Antonio, Texas. See the website (<http://ncue.tamu.edu>) for deadline dates and updated information. February 17th is the deadline for paper submissions and the hotel registration deadline is April 21, 2014.

Livestock Insect Worker Conference

The 2014 Livestock Insect Worker Conference will be held in San Diego, CA from June 22-25, 2014. The conference will be held at the Bahia Resort on Mission Bay (<http://www.bahiahotel.com/>) providing a beachfront venue for the conference. A website for conference registration and hotel reservations will be available starting in January 2014. However, this announcement is intended to provide the conference dates so that you can put the event on your long-range calendars and plan to attend. We hope that most of you will have the opportunity to join us in California. The hotel room rates are an exceptional deal starting at just \$149/night (for standard room) and we have negotiated to have this rate available from the weekend before the Conference through the end of the weekend following the conference.

If you are unfamiliar with the Livestock Insect Worker Conference (LIWC), this will be the 58th Annual Meeting of this organization focused on research and extension related to the management of arthropod pests of animals. The Conference began with a

1956 meeting in Kerrville, TX focused on systemic insecticides for use on livestock. The Conference is organized into symposia by animal commodity (cattle, poultry, sheep, horses, etc...) providing university and public agency researchers and private industry representatives to present recent work, knowledge, or ideas.

A call for presentations (and posters) will be sent out in mid-winter. In addition, the LIWC sponsors two annual awards: 1) The Lifetime Achievement Award to a university or public agency entomologist who has made significant contributions to the field of livestock entomology, and 2) The Industry Appreciation Award to a representative from one of the LIWC industry partners in recognition of contributions to animal health and productivity.

If you have questions about the conference, you may forward them to the 2014 LIWC Local Arrangements Co-Chair, Dr. Alec Gerry (see contact info below). We look forward to having you participate in this conference. Watch for more information on the Conference to be provided in early-mid winter.

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Research in Progress

"Systematics, Coevolution, and Microbial Community Structure of Bat Fleas (Siphonaptera: Ischnopsyllidae)"

Principal Investigator: Katharina Dittmar De La Cruz, kd52@buffalo.edu
Organization: SUNY at Buffalo, 402 Crofts Hall, Buffalo, NY 14260

Bats constitute nearly 25% of the diversity of all mammals. They have a global distribution, and are key players in diverse ecosystems. Like any other mammal, they are exposed to, and suffer from, a range of parasites. Among these parasites are a group of bloodsucking insects - the bat fleas. Contrary to many other fleas, they are uniquely adapted to bats. Currently, even basic knowledge about these parasites is scarce. This grant sets out to remedy this problem by pursuing the following important goals: 1) for the first time comprehensive information about the development, diversity, global distribution, host association, and ecology of these fleas will be gathered, 2) the evolutionary history of bat fleas and the timing of their diversification will be explored, in the context of parallel evolution with their hosts, and 3) the previously unknown

microbial diversity of bat fleas will be studied and linked across different parasite developmental stages and ecosystems, characterizing beneficial as well as potentially pathogenic bacteria.

The results of these interdisciplinary studies will benefit a wide range of researchers (e.g. systematists, microbiologists, conservation biologists, parasitologists, epidemiologists). The proposed studies will enhance current knowledge about symbiosis, parasitism, co-evolution, and disease ecology, and will form the foundation for further research in these fields. This project will provide educational opportunities for graduate and undergraduate student training in important research areas, such as parasitology, metagenomics, and phylogenetics. Minority undergraduates will be especially recruited, and academic success of these students will be monitored by a novel mentoring approach, called "Research in Pairs". This approach centers on teaching students the value of ethics, communication, and mentoring while they are pursuing their own research projects.

New Books

Biology of Ticks, Volumes I and II, Second Edition.
Daniel E. Sonenshine and R. Michael Roe.

Biology of Ticks is the most comprehensive work on tick biology and tick-borne diseases. This second edition is a multi-authored work, featuring the research and analyses of renowned experts across the globe. Spanning two volumes, the book examines the systematics, biology, structure, ecological adaptations, evolution, genomics, and the molecular processes that underpin the growth, development, and survival of these important disease-transmitting parasites. Also discussed is the remarkable array of diseases transmitted (or caused) by ticks, as well as modern methods for their control. This book should serve as a modern reference for students, scientists, physicians, veterinarians and other specialists.

Volume I covers the biology of the tick and features chapters on tick systematics, tick life cycles, external and internal anatomy, and others dedicated to specific organ systems, specifically, the tick integument, mouthparts and digestive system, salivary glands, waste removal, respiratory system, circulatory system and hemolymph, fat body, the nervous and sensory systems and reproductive systems. Nov 2013 • 560 pp. • Hardcover 9780199744053, \$120.00.

Volume II includes chapters on the ecology of non-nidicolous and nidicolous ticks, genetics, and genomics (including the genome of the Lyme disease vector *Ixodes scapularis*) and immunity, including host immune responses to tick feeding and tick-host interactions, as well as the tick's innate immune system that prevents and/or controls microbial infections. Dec 2013 • 496 pp. • Hardcover 9780199744060, \$120.00. (Review from Oxford University Press.)

Featured research

ACOSTA, ROXANA & JUAN J. MORRONE. 2013. Phylogenetics of the tribe Phalacropsyllini (Siphonaptera: Ctenophthalmidae: Neopsyllinae) based on molecular and morphological evidence. *Zootaxa* 3630 (2): 333–346.

Six genera of the tribe Phalacropsyllini were analysed based on molecular (18S rDNA and 28S rDNA) and morphological characters, using maximum parsimony analyses and Bayesian inference. The results support the paraphyly of the tribe Phalacropsyllini, as shown in a previous study based only on molecular characters. Three main clades were identified: one includes the genera *Neopsylla*, *Epitedia*, *Catallagia*, and *Delotelis*; another includes *Phalacropsylla* and *Strepsylla*; and a third one includes only *Meringis*. In order to propose a phylogenetic classification we recognize three tribes, transferring *Epitedia*, *Catallagia* and *Delotelis* to Neopsyllini; keeping Phalacropsyllini only for *Phalacropsylla* and *Strepsylla*; and proposing the new tribe Meringini for *Meringis*.

Acosta-Gutiérrez, R. 2013. Biodiversidad de Siphonaptera en México. *Revista Mexicana de Biodiversidad* 84 DOI: 10.7550/rmb.35267.

Siphonaptera are insect parasites of endotherm vertebrates, birds and mammals, occurring more abundantly in the Order Rodentia. Fleas are distributed widely in the world, except in Antarctica [see Editor's note below], and are considered cosmopolitan. This group of insects is diverse, with 2,575 species reported worldwide. Mexico has 172 species that belong to 8 families; representing 6.8% of the world's flea fauna. Ceratophyllidae (74 species) and Ctenophthalmidae (45 species) are the most abundant families in Mexico. Fleas have sanitary importance because they are vectors of diseases like plague, typhus and helminthiasis.

[Editor's note: Fleas are in Antarctica, too. *Glaciopsyllus antarcticus*!]

American Chemical Society (2013, September 12). Sewage treatment removes widely used home and garden insecticides from wastewater. *ScienceDaily*. Retrieved November 7, 2013, from <http://www.sciencedaily.com/releases/2013/09/130912143622.htm>

Even though sewage treatment plants are not designed to remove tiny amounts of pesticides, they do an excellent job of dealing with the most widely used family of home and garden insecticides, scientists have reported. Their study on pyrethroid insecticides -- used in more than 3,500 products -- was part of the 246th National Meeting & Exposition of the American Chemical Society (ACS), held in Indianapolis this week.

"We found that advanced sewage treatment reduced the levels of pyrethroids by more than 97 percent," said Kurt N. Ohlinger, Ph.D., who presented the results of the study. "That's a reduction to less than one part per trillion, and pretty impressive."

Conventional treatment processes remove about 89 percent, something we knew from previous research. Based on these results, we do not expect the trace amounts of pyrethroids in sewage treatment plant effluent to be toxic to even the most sensitive aquatic life."

Use of pyrethrins, derived from chrysanthemum flowers, and the related synthetic pyrethroids, has been on the increase during the last decade. They are replacing organophosphate pesticides, which are more acutely toxic to birds and mammals, with uses that include home insect control, insect-repellant clothing, dog and cat flea shampoos, mosquito control and agriculture.

The growing use led Ohlinger and colleagues to check on the effectiveness of advanced sewage treatment processes in removing pyrethroids from wastewater from a sewage treatment plant. They are with the Sacramento Regional County Sanitation District in California, which was implementing advanced sewage treatment.

Beaucournu J.-C., Belaz S., Munoz-Leal S. & Gonzalez-Acuna D. 2013. A new flea, *Ectinorus (Ectinorus) insignis* n. sp. (Siphonaptera, Rhopalopsyllidae, Parapsyllinae), with notes on the subgenus *Ectinorus* in Chile and comments on unciform sclerotization in the superfamily Malacopsylloidea. *Parasite*, 2013, 20, 35. DOI : 10.1051/parasite/2013033

Basra, Gurjot, Megan A. Berman, and Lucas S. Blanton. 2012. Murine Typhus: An Important Consideration for the Nonspecific Febrile Illness. *Case Reports in Medicine* Volume 2012, Article ID 134601, 3 pages. doi:10.1155/2012/134601

Murine typhus is a widely distributed flea-borne infection caused by *Rickettsia typhi*. Symptoms of murine typhus are nonspecific and mimic a variety of other infectious diseases. We herein report a case of murine typhus in an area where the broad use of DDT in the mid-20th century has now made it a rare disease. The patient described presented with headache, fever, and a faint macular rash. Initial laboratory studies revealed a slight transaminase elevation. Further questioning revealed exposure to opossums, prompting the consideration of murine typhus as a diagnosis. Although typhus group antibodies were not present during the patient's acute illness, empiric therapy with doxycycline was initiated, and the patient defervesced. One month after convalescence, the patient returned to clinic with serum that contained typhus group antibodies with an IgG titer of 1 : 1024. Murine typhus is an important consideration during the workup of a patient with a nonspecific febrile illness. Exposure to reservoir hosts and the flea vector place humans at risk for this disease. Clinician recognition of this entity is required for diagnosis and effective therapy.

Chandler JA, James PM (2013) Discovery of Trypanosomatid Parasites in Globally Distributed *Drosophila* Species. *PLoS ONE* 8(4): e61937. doi:10.1371/journal.pone.0061937

Microbial parasites of animals include bacteria, viruses, and various unicellular eukaryotes. Because of the difficulty in studying these microorganisms in both humans and disease vectors, laboratory models are commonly used for experimental analysis of host-parasite interactions. *Drosophila* is one such model that has made significant contributions to our knowledge of bacterial, fungal, and viral infections. Despite this, less is known about other potential parasites associated with natural *Drosophila* populations. Here, we surveyed sixteen *Drosophila* populations comprising thirteen species from four continents and Hawaii and found that they are associated with an extensive diversity of trypanosomatids (Euglenozoa, Kinetoplastea). Phylogenetic analysis finds that *Drosophila*-associated trypanosomatids are closely related to taxa that are responsible for various types of leishmaniasis and more distantly related to the taxa responsible for human African trypanosomiasis and Chagas disease. We suggest that *Drosophila* may provide a powerful system for studying the interactions between trypanosomatids and their hosts.

[editor's extended extract]: "Monoxenous trypanosomatids have been detected in Diptera, Hemiptera, Hymenoptera, Lepidoptera, and Siphonaptera ... although the true diversity of insect-associated trypanosomatids is likely far from realized.... Indeed, extensive surveys within Heteroptera have found that nearly a quarter of all individuals are infected, many with previously undescribed trypanosomatid strains These surveys challenge the “one host – one parasite” view of trypanosomatid infection ... and suggest that the factors defining host-parasite specificity are not well understood in monoxenous trypanosomatids."

Coles TB, Dryden MW. A review of insecticide/acaricide resistance in fleas and ticks infesting dogs and cats. *Parasites & Vectors* (in press).

Dryden MW, Payne PA, Smith V, Chwala M, Jones E, Davenport J, Fadl G, Martinez-Perez de Zeiders MF, Heaney K, Ford P, Sun F. Evaluation of indoxacarb and fipronil (s)-methoprene topical spot-on formulations to control flea populations in naturally infested dogs and cats in private residences in Tampa FL. USA. *Parasites & Vectors* (in press).

Darries-Vallier, Anne, Aurélien Ausset and Pierre Besrest. 2013. Experimental Implantation Trials of *Xenopsylla cunicularis* Smit (Siphonaptera: Pulicidae) in Northern France with the Objective to Use it as Vaccine Vector. *International Journal of Insect Science* 2013:5 21–34. doi: 10.4137/IJIS.S11804

To combat animal diseases, we propose that *Xenopsylla cunicularis* Smit (Siphonaptera: Pulicidae), a specific flea of the European wild rabbit *Oryctolagus cuniculus* L. (Lagomorpha), can be used to carry a vaccine into wild rabbit populations to protect them against lethal diseases. *Oryctolagus cuniculus* is widespread throughout Europe, but *X. cunicularis* occurs naturally only in drier areas of Morocco, Spain, and southwestern France, raising questions about the flea's general use and the subsequent

risk of uncontrolled proliferation outside its natural distribution.

To evaluate this risk, fleas were released in five experimental enclosures containing rabbits (four in northern France and one in southwestern France as a control). Approximately one year later, adult and immature fleas were recovered from rabbits and warrens. The climate during the experiments was recorded and warren substrate granulometry was defined.

Our results showed that northern France is not suitable for persistence of *X. cunicularis* because low temperatures reduce flea development and high rainfall all over the year keeps the soil damp, which is asphyxic for fleas, even on a sandy substrate. These implantation trials suggest that uncontrolled proliferation and permanent establishment of fleas are unlikely in northern France.

Darvishi, M. M., M. R. Youssefi, E. Changizi, R. R. Lima, and M. T. Rahimi. 2014. A New Flea from Iran. *Asian Pacific Journal of Tropical Disease* 4 (2): 85-87.

De Avelar, D. M. D., Filho, E. J. F., & Linardi, P. M. (2013). A new species of *Tunga* (Siphonaptera: Tungidae) parasitizing cattle from Brazil. *Journal of medical entomology*, 50(4), 679-684.

Linardi, P. M., De Avelar, D. M., & Facury Filho, E. J. (2013). Establishment of *Tunga trimamillata* (Siphonaptera: Tungidae) in Brazil. *Parasitology research*, 112(9), 3239-3242.

Tunga trimamillata is a species of sand flea occurring in Ecuador and Peru parasitizing cattle, goat, sheep, swine, and man. This is the first report of this species in Brazil, having been found on the hooves of cows in Barretos, São Paulo State, and Felixlândia, Minas Gerais State, and previously misidentified as *Tunga penetrans*. A previous report concerning *Hydrochoerus hydrochaeris* from Rio Novo, Minas Gerais State, may also be attributed to that species of sand flea, a possible the primary host. Given the large geographical distribution of *T. trimamillata*, the vast cattle population in Brazil, and the high number of individuals subject to the risk of tungiasis, the number of cases attributed to this sand flea will most likely increase over time.

Linardi, P. M. 2013. Os Ectoparasitos de Marsupiais Brasileiros. In: N. C. Cáceres (ed.) Os Marsupiais do Brasil. Biologia, Ecologia e Conservação. 2a. edição. Editora UFMS, Campo Grande, Brasil. p. 127-155.

Linardi, P. M. Checklist dos Siphonaptera do Estado do Mato Grosso do Sul. Biota Neotropica (in press).

DI IORIO, OSVALDO & PAOLA TURIENZO. 2011. A preliminary bibliographic survey of the insects found in poultry houses from the Neotropical Region, with remarks on selected taxa shared with native birds' nests. *Zootaxa* 2858: 1–60.

Species of insects associated to the habitat of *Gallus gallus* (Aves: Phasianidae) in the Neotropical Region belong to 144 identified species (42 Coleoptera; 14 Diptera; 17 Hymenoptera; 9 Siphonaptera; 1 Lepidoptera; 56 Hemiptera (one hybrid); 5 Dermaptera); 33 identified to genus (21 Coleoptera; 3 Diptera; 7 Hymenoptera; 1 Hemiptera; 1 Blattaria); 37 identified to family (23 Coleoptera; 9 Diptera; 2 Hymenoptera; 1 Lepidoptera; 2 Hemiptera); and 6 to order (2 Coleoptera; 1 Hymenoptera; 1 Siphonaptera; 1 Psocoptera; 1 Dermaptera). Most of the insects are haematophagous ectoparasites (Hemiptera; Siphonaptera; Diptera), detritivores (Coleoptera; Lepidoptera; Diptera; Blattaria), predators (Coleoptera; Diptera; Hemiptera; Dermaptera; Hymenoptera), and parasitoids (Hymenoptera). A total of 46 native American species and/or subspecies of Triatominae (Hemiptera: Reduviidae) were found in chicken houses, from which 18 were also found in birds' nests. It was recently observed that other insects from poultry houses, generally exotic species, had colonized native American birds' nests. Comments and remarks of selected taxa also found in birds' nests from Argentina and other countries are provided.

Durden, L. A., & Richardson, D. J. (2013). Ectoparasites of the Virginia opossum (*Didelphis virginiana*), Raccoon (*Procyon lotor*), and Striped skunk (*Mephitis mephitis*) from Keith county, Nebraska. Transactions of the Nebraska Academy of Sciences 33, 21–24.

Six Virginia opossums (*Didelphis virginiana*), nine raccoons (*Procyon lotor*) and one striped skunk (*Mephitis mephitis*) collected from Keith County Nebraska were examined for ectoparasites. All three host species were parasitized by adults of the American dog tick, *Dermacentor variabilis*. Opossums were also parasitized by the flea *Pulex simulans* and the tiny fur mite *Didelphilichus serratifer*; the latter species represents a new state record for Nebraska. Raccoons were also parasitized by *P. simulans* and by the lagomorph-associated flea *Euhoplopsyllus glacialis affinis*, whereas the skunk was also parasitized by the chewing louse *Neotrichodectes mephiditis*.

Gascuel F, Choisy M, Duplantier J-M, Débarre F, Brouat C (2013) Host Resistance, Population Structure and the Long-Term Persistence of Bubonic Plague: Contributions of a Modelling Approach in the Malagasy Focus. PLoS Comput Biol 9(5): e1003039. doi:10.1371/journal.pcbi.1003039

Although bubonic plague is an endemic zoonosis in many countries around the world, the factors responsible for the persistence of this highly virulent disease remain poorly known. Classically, the endemic persistence of plague is suspected to be due to the coexistence of plague resistant and plague susceptible rodents in natural foci, and/or to a metapopulation structure of reservoirs.

Here, we test separately the effect of each of these factors on the long-term

persistence of plague. We analyse the dynamics and equilibria of a model of plague propagation, consistent with plague ecology in Madagascar, a major focus where this disease is endemic since the 1920s in central highlands.

By combining deterministic and stochastic analyses of this model, and including sensitivity analyses, we show that (i) endemicity is favoured by intermediate host population sizes, (ii) in large host populations, the presence of resistant rats is sufficient to explain long-term persistence of plague, and (iii) the metapopulation structure of susceptible host populations alone can also account for plague endemicity, thanks to both subdivision and the subsequent reduction in the size of subpopulations, and extinction-recolonization dynamics of the disease. In the light of these results, we suggest scenarios to explain the localized presence of plague in Madagascar.

Bubonic plague, known to have marked human history by three deadly pandemics, is an infectious disease which mainly circulates in wild rodent populations and is transmitted by fleas. Although this disease can be quickly lethal to its host, it has persisted on long-term in many rodent populations around the world.

The reasons for this persistence remain poorly known. Two mechanisms have been invoked, but not yet explicitly and independently tested: first, the spatial structure of rodent populations (subdivision into several subpopulations) and secondly, the presence of, not only plague-susceptible rodents, but also plague-resistant ones. To gain insight into the role of the above two factors in plague persistence, we analysed a mathematical model of plague propagation. We applied our analyses to the case of Madagascar, where plague has persisted on central highlands since the 1920s and is responsible for about 30% of the human cases worldwide. We found that the long-term persistence of plague can be explained by the presence of any of the above two factors. These results allowed us to propose scenarios to explain the localized presence of plague in the Malagasy highlands, and help understand the persistence of plague in many wild foci.

GÓMEZ, M. S., and SHIMALOV, V. 2013. Fleas found on small mammals in drainage channels in two districts of South-West Belarus. *Rev. Ibero-Latinoam. Parasitol.* 72 (1): 61-67.

This article discusses fleas detected among 196 Cricetidae rodents (2 Arvicolinae species and 4 Murinae species) living in five different biotopes located along the banks of drainage channels in South-West Belarus (Brest and Malorita districts). Twenty six small mammals representing four different species were found to be parasitized by fleas (13.3%). Neither *Apodemus flavicollis* nor *Mus musculus* harboured siphonaptera. A total of fifty nine fleas have been detected. The flea species were *Hystriehopsylla orientalis orientalis*, *Ctenophthalmus (Ctenophthalmus) agyrtes kleinschmidtianus*, *Ctenophthalmus (Euctenophthalmus) assimilis assimilis*, *Megabothris (Gebiella) turbidus* and *Nosopsyllus (Nosopsyllus) fasciatus*. *C. (E.) assimilis assimilis* proved to be the most abundant flea species, the most prevalent and also the most widely-spread. Results show that biotopes with apparently analogous characteristics do not share similar flea biocenosis.

Govender, Thashlin, Jo M. Barnes, and Clarissa H. Pieper. 2011. The Impact of Densification by Means of Informal Shacks in the Backyards of Low-Cost Houses on the Environment and Service Delivery in Cape Town, South Africa. *Environmental Health Insights* 5: 23–52. doi: 10.4137/EHI.S7112

This paper investigates the state-sponsored low cost housing provided to previously disadvantaged communities in the City of Cape Town. The strain imposed on municipal services by informal densification of unofficial backyard shacks was found to create unintended public health risks. Four subsidized low-cost housing communities were selected within the City of Cape Town in this cross-sectional survey. Data was obtained from 1080 persons with a response rate of 100%. Illegal electrical connections to backyard shacks that are made of flimsy materials posed increased fire risks. A high proportion of main house owners did not pay for water but sold water to backyard dwellers. The design of state-subsidised houses and the unplanned housing in the backyard added enormous pressure on the existing municipal infrastructure and the environment. Municipal water and sewerage systems and solid waste disposal cannot cope with the increased population density and poor sanitation behaviour of the inhabitants of these settlements. The low-cost housing program in South Africa requires improved management and prudent policies to cope with the densification of state-funded low-cost housing settlements.

[Editor's extended extract]: "All of the respondents in the survey complained of pests carrying potential health risks within their immediate home environment. The respondents from all four communities reported their most prominent pest problem was rats by 50%, cockroaches by 30%, fleas by 16% and flies by 4%. Flies were observed in all homes, indicating the presence of flies were underreported."

Graham SP, Hassan HK, Burkett-Cadena ND, Guyer C, Unnasch TR (2009) Nestedness of Ectoparasite-Vertebrate Host Networks. *PLoS ONE* 4(11): e7873. doi:10.1371/journal.pone.0007873

Determining the structure of ectoparasite-host networks will enable disease ecologists to better understand and predict the spread of vector-borne diseases. If these networks have consistent properties, then studying the structure of well-understood networks could lead to extrapolation of these properties to others, including those that support emerging pathogens. Borrowing a quantitative measure of network structure from studies of mutualistic relationships between plants and their pollinators, we analyzed 29 ectoparasite-vertebrate host networks—including three derived from molecular bloodmeal analysis of mosquito feeding patterns—using measures of nestedness to identify non-random interactions among species.

We found significant nestedness in ectoparasite-vertebrate host lists for habitats ranging from tropical rainforests to polar environments. These networks showed non-random patterns of nesting, and did not differ significantly from published estimates of nestedness from mutualistic networks. Mutualistic and antagonistic networks appear to be organized similarly, with generalized ectoparasites interacting with hosts that attract

many ectoparasites and more specialized ectoparasites usually interacting with these same “generalized” hosts. This finding has implications for understanding the network dynamics of vector-borne pathogens. We suggest that nestedness (rather than random ectoparasite-host associations) can allow rapid transfer of pathogens throughout a network, and expand upon such concepts as the dilution effect, bridge vectors, and host switching in the context of nested ectoparasite-vertebrate host networks.

[editor's extended abstract]: "Our results have implications for understanding disease transmission cycles. In the specific case of mosquitoes at Tuskegee National Forest (TNF), nesting assures that novel diseases, such as West Nile virus, have the potential to spread widely among vertebrate hosts. This prediction is associated with the fact that nesting results from a core ectoparasite (*Culex erraticus*) [see Editor's note below.] that feeds on the majority of vertebrate hosts and a core host (White-tailed deer) that is fed upon by the majority of mosquitoes. These core taxa greatly reduce the number of linkages needed to unite all taxa within the network. Thus, only six mosquitoes are required to retain all interactions of the TNF network whereas 15 mosquitoes are required for a randomized version of the same network. The concept of compartmentalized nestedness predicts that pathogens likely coevolve with hosts and ectoparasites within specific compartments, but the possibility for escape into additional or all other compartments is possible due to the high connectance of the network. Zoonoses (diseases caused by pathogens which normally cycle through wildlife) can thus be the result of compartment-spanning or compartment-shifting pathogens, and network analysis may lead to discovery of likely conditions that exacerbate such phenomena (e.g., during peaks of host or ectoparasite abundance)."

[Editor's note: Are mosquitoes ectoparasites?]

Gao, T. -P, C. -K Shih, X. Xu, S. Wang, and D. Ren. 2012. Mid-Mesozoic Flea-Like Ectoparasites of Feathered Or Haired Vertebrates. *Current Biology* 22 (8): 732-735.

Fleas are a group of highly specialized blood-feeding ectoparasites whose early evolutionary history is poorly known Although several recent discoveries have shed new light on the origin of the group ..., a considerable gap exists between stem fleas and crown fleas. Here we report a new transitional flea, *Saurophthirus exquisitus* sp. nov., assigned to a new family Saurophthiridae fam. nov., from the Lower Cretaceous Yixian Formation of northeastern China. Saurophthirids are more similar to crown fleas than other stem fleas in having a relatively small body size, relatively short and slender piercing-sucking stylet mouthparts, comparably short and compact antennae, rows of short and stiff bristles on the thorax, and highly elongated legs. The new finding greatly improves our understanding of the morphological transition to the highly specialized body plan of extant fleas. However, saurophthirids also display several features unknown in other fleas, and some of these features are suggestive of a possible ectoparasitic relationship to contemporaneous pterosaurs, though other possibilities exist. The new fossils, in conjunction with previous discoveries, highlight a broad diversity of

ectoparasitic insects in the mid-Mesozoic.

Huang, D., M. S. Engel, C. Cai, H. Wu, and A. Nel. 2012. Diverse Transitional Giant Fleas from the Mesozoic Era of China. *Nature* 483 (7388): 201-204.

Fleas are one of the major lineages of ectoparasitic insects and are now highly specialized for feeding on the blood of birds or mammals ...This has isolated them among holometabolous insect orders, although they derive from the Antliophora (scorpionflies and true flies). Like most ectoparasitic lineages, their fossil record is meagre and confined to Cenozoic-era representatives of modern families..., so that we lack evidence of the origins of fleas in the Mesozoic era. The origins of the first recognized Cretaceous stem-group flea, *Tarwinia*, remains highly controversial....

Here we report fossils of the oldest definitive fleas—giant forms from the Middle Jurassic and Early Cretaceous periods of China. They exhibit many defining features of fleas but retain primitive traits such as non-jumping hindlegs. More importantly, all have stout and elongate sucking siphons for piercing the hides of their hosts, implying that these fleas may be rooted among the pollinating ‘long siphonate’ scorpionflies of the Mesozoic. Their special morphology suggests that their earliest hosts were hairy or feathered ‘reptilians’, and that they radiated to mammalian and bird hosts later in the Cenozoic.

Huang, D., M. S. Engel, C. Cai, and A. Nel. 2013. Mesozoic giant fleas from northeastern China (Siphonaptera): Taxonomy and implications for palaeodiversity. *Chinese Science Bulletin*, 2013, 58: 1682–1690, doi: 10.1007/s11434-013-5769-3

The recently discovered definite giant fleas from the Middle Jurassic Daohugou fauna and the Early Cretaceous Jehol fauna of northeastern China represent significant evidence for understanding ectoparasitism in the Mesozoic as well as the evolution of these giant blood feeders with their putative hosts (i.e. hairy or feathered vertebrates). On the basis of seven well-preserved specimens from Daohugou and Huangbanjigou we analyse the systematic classification of these primitive fleas, establishing two new genera and three new species as *Pseudopulex wangi* sp. nov., *Hadropsylla sinica* gen. et sp. nov., and *Tyrannopsylla beipiaoensis* gen. et sp. nov. All of them are assigned to the extinct siphonapteran family Pseudopulicidae, while the Early Cretaceous genus *Tarwinia* is transferred to Tarwiniidae fam. nov. The basal morphological disparities of Siphonaptera in the Mesozoic are evidenced by the occurrence of at least three distinct groups (pseudopulicids, tarwiniids, and saurophthirids). These disparate morphologies likely indicate adaptations to different hosts.

Harbeck M, Seifert L, Hänsch S, Wagner DM, Birdsell D, et al. (2013) *Yersinia pestis* DNA from Skeletal Remains from the 6th Century AD Reveals Insights into Justinianic Plague. *PLoS Pathog* 9(5): e1003349. doi:10.1371/journal.ppat.1003349

Yersinia pestis, the etiologic agent of the disease plague, has been implicated in

three historical pandemics. These include the third pandemic of the 19th and 20th centuries, during which plague was spread around the world, and the second pandemic of the 14th–17th centuries, which included the infamous epidemic known as the Black Death. Previous studies have confirmed that *Y. pestis* caused these two more recent pandemics. However, a highly spirited debate still continues as to whether *Y. pestis* caused the so-called Justinianic Plague of the 6th–8th centuries AD. By analyzing ancient DNA in two independent ancient DNA laboratories, we confirmed unambiguously the presence of *Y. pestis* DNA in human skeletal remains from an Early Medieval cemetery. In addition, we narrowed the phylogenetic position of the responsible strain down to major branch 0 on the *Y. pestis* phylogeny, specifically between nodes N03 and N05. Our findings confirm that *Y. pestis* was responsible for the Justinianic Plague, which should end the controversy regarding the etiology of this pandemic. The first genotype of a *Y. pestis* strain that caused the Late Antique plague provides important information about the history of the plague bacillus and suggests that the first pandemic also originated in Asia, similar to the other two plague pandemics.

Author Summary: Plague is a notorious and fatal human disease caused by the bacterium *Yersinia pestis* that is endemic in many countries around the world. Three of the most devastating pandemics in human history have been associated with plague. The second pandemic originated in central Asia and peaked in Europe between 1348 and 1350 (a period of time known as the Black Death). The third pandemic began in the Yunnan province of China in the mid-1850s and subsequently spread to Africa, the Americas, Australia, Europe, and other parts of Asia. The second and third pandemics are well documented and scientifically proven. However, the first pandemic, which began in the 6th century and is also known as Justinianic Plague, is still a matter of controversy. Recently it has been suggested that Justinian's plague was not caused by *Y. pestis*. We detected *Y. pestis* DNA in samples obtained from multiple 6th century skeletons from Germany. This confirms that Justinianic Plague crossed the Alps and affected local populations there, including current day Bavaria. Furthermore, we used DNA fingerprinting approaches to determine Asia as the likely geographic origin for these strains.

Hayashiya, S. et al. 2012. Infestation by the cat flea *Ctenocephalides felis* showing low sensitivity to imidacloprid in a dog. *Jpn. J. Vet Dermatol.* 18 (2): 93-98.

Horspool, L. J. (2013). Animal Health Markets and Opportunities: Companion Animal Landscape. in Long Acting Animal Health Drug Products (pp. 15-46), Animal Health Markets and Opportunities: Companion Animal Landscape, Rathbone, Michael J.; McDowell, Arlene (Eds.), XII, 389 p. 68 illus., 50 illus. in color. Springer US.

In the future, the global market for companion animal health products is expected to further grow and become more specialized. The major drivers will be the continued strengthening of the bond between owners and their animal companions, increasing companion animal owner awareness, and increasing companion animal owner demands and expectations for companion animal care. This chapter discusses the background

knowledge that is required to enable a formulator to differentiate their drug products through novel delivery technologies which address the future companion animal global market needs with the goal of increasing companion animal owner compliance and thus enhancing efficacy.

Khokhlova, I. S., Pilosof, S., Fielden, L. J., Degen, A. A., & Krasnov, B. R. (2013). A trade-off between quantity and quality of offspring in haematophagous ectoparasites: the effect of the level of specialization. *Journal of Animal Ecology*.

Theory predicts an adaptive trade-off between quantity and quality of offspring if mothers can reliably predict the offspring environment. We studied egg production and quality of offspring in two flea species (host-specialist *Parapulex chephrenis* and host-generalist *Xenopsylla ramesis*) exploiting eight rodent species. We evaluated quality of new imagoes via their developmental time, size (length of a femur as a proxy) and resistance to starvation without a blood meal.

We predicted that the offspring quality would increase with (i) a decrease in the number of eggs produced by mothers and (ii) an increase in phylogenetic distance between maternal host and principal host of a flea. We also predicted that negative relationships between offspring quality and either maternal egg production effort or phylogenetic distance between maternal host and the principal host or both would be manifested stronger in host-opportunistic than in host-specific fleas.

The highest number of eggs produced per female flea was accompanied by the longest duration of development and the smallest offspring in *X. ramesis*, while *P. chephrenis* that hatched from larger clutches survived for less time under starvation. Although there was no significant effect of host species on any dependent variable, association between offspring quality and phylogenetic distance of the maternal host from the principal host of a flea was found in *X. ramesis* (but not *P. chephrenis*) with new imagoes being larger if their maternal hosts were phylogenetically distant from the principal host. Our results demonstrated stronger trade-off between quantity and quality of offspring in a generalist than in a specialist flea, supporting the association between life-history plasticity and generalist feeding strategy.

Krasnov, B. R., Pilosof, S., Shenbrot, G. I., & Khokhlova, I. S. (2013). Spatial variation in the phylogenetic structure of flea assemblages across geographic ranges of small mammalian hosts in the Palearctic. *International journal for parasitology*. 43 (9): 763-770.

We investigated spatial variation in the phylogenetic structure (measured as a degree of phylogenetic clustering) of flea assemblages across the geographic ranges of 11 Palearctic species of small mammalian hosts and asked whether the phylogenetic structure of the flea assemblage of a host in a locality is affected by (i) distance of this locality from the centre of the host's geographic range, (ii) geographic position of the locality (distance to the equator) and (iii) phylogenetic structure of the entire flea assemblage of the locality. Our results demonstrated that the key factor underlying spatial

variation of the phylogenetic structure of the flea assemblage of a host was the distance from the centre of the host's geographic range. However, the pattern of this spatial variation differed between host species and might be explained by their species-specific immunogenetic and/or distributional patterns. Local flea assemblages may also, to some extent, be shaped by environmental filtering coupled with historical events. In addition, the phylogenetic structure of a local within-host flea assemblage may mirror the phylogenetic structure of the entire across-host flea assemblage in that locality and, thus, be affected by the availability of certain phylogenetic lineages.

Krasnov, B. R., Pilosof, S., Stanko, M., Morand, S., Korralo-Vinarskaya, N. P., Vinarski, M. V., & Poulin, R. (2013). Co-occurrence and phylogenetic distance in communities of mammalian ectoparasites: limiting similarity versus environmental filtering. *Oikos* 123 (1): 63-70.

Similarity between species plays a key role in the processes governing community assembly. The co-occurrence of highly similar species may be unlikely if their similar needs lead to intense competition (limiting similarity). On the other hand, persistence in a particular habitat may require certain traits, such that communities end up consisting of species sharing the same traits (environmental filtering). Relatively little information exists on the relative importance of these processes in structuring parasite communities.

Assuming that phylogenetic relatedness reflects ecological similarity, we tested whether the co-occurrence of pairs of flea species (Siphonaptera) on the same host individuals was explained by the phylogenetic distance between them, among 40 different samples of mammalian hosts (rodents and shrews) from different species, areas or seasons. Our results indicate that frequency of co-occurrence between flea species increased with decreasing phylogenetic distance between them in 37 out of 40 community samples, with 14 of these correlations being statistically significant. A meta-analysis across all samples confirmed the overall trend for closely related species to co-occur more frequently on the same individual hosts than expected by chance, independently of the identity of the host species or of environmental conditions. These findings suggest that competition between closely related, and therefore presumably ecologically similar, species is not important in shaping flea communities. Instead, if only fleas with certain behavioural, ecological and physiological properties can encounter and exploit a given host, and if phylogenetic relationships determine trait similarity among flea species, then a process akin to environmental filtering, or host filtering, could favour the co-occurrence of related species on the same host.

Pilosof, S., Fortuna, M. A., Vinarski, M. V., Korralo-Vinarskaya, N. P., & Krasnov, B. R. (2013). Temporal dynamics of direct reciprocal and indirect effects in a host-parasite network. *Journal of Animal Ecology*. 82 (5): 987-996.

Temporal variation in the direct and indirect influence that hosts and parasites exert on each other is still poorly understood. However, variation in species' influence due to species and interactions turnover can have important consequences for host community dynamics and/or for parasite transmission dynamics, and eventually for the risk of zoonotic diseases. We used data on a network of small mammals and their

ectoparasites surveyed over 6 years to test hypotheses exploring (i) the temporal variability in direct and indirect influences species exert on each other in a community, and (ii) the differences in temporal variability of direct/indirect influences between temporally persistent (TP) and temporally intermittent species. We modelled the temporal variation in (i) direct reciprocal influence between hosts and parasites (hosts providing resources to parasites and parasites exploiting the resources of hosts), using an asymmetry index, and (ii) indirect influence among species within a community (e.g. facilitation of parasite infestation by other parasites), using betweenness centrality. We also correlated asymmetry and centrality to examine the relationship between them. Network dynamics was determined by TP species but even those species had strong among-species heterogeneity in the temporal variation of the direct/indirect effects they exerted. In addition, there was a significant positive linear correlation between asymmetry and centrality. We conclude that the temporal dynamics of host–parasite interactions is driven by TP hosts. However, even within this group of persistent species, some exhibit large temporal variation, such that the functional roles they play (e.g. in promoting parasite transmission) change over time. In addition, parasites having a large negative impact on hosts are also those facilitating the spread of other parasites through the entire host community. Our results provide new insights into community dynamics and can be applied in the management of antagonistic networks aimed at preventing disease outbreaks.

Kraft, J. P., & Stapp, P. (2013). Movements and burrow use by northern grasshopper mice as a possible mechanism of plague spread in prairie dog colonies. *Journal of Mammalogy*. 94 (5): 1087-1093.

Northern grasshopper mice (*Onychomys leucogaster*) have been implicated as a potential reservoir for plague, which causes local extinction of black-tailed prairie dog (*Cynomys ludovicianus*) colonies. To identify mechanisms by which grasshopper mice could facilitate plague spread, we measured burrow use, movements, and flea burdens of mice in colonies in northern Colorado. At the microhabitat scale, powder-tracked mice ($n = 41$) used both prairie dog mounds and burrows extensively, entering an average of 5.3 burrows per 100 m traveled. Burrow use did not differ between active and inactive mounds, or vary with mouse age, sex, or reproductive status. Radiotracking revealed that mice occupied larger ranges ($\bar{X} = 3.84$ ha) than reported off colonies, which we estimated would overlap 12–23 prairie dog coterie territories. Mice also harbored high flea burdens (8.1 fleas/mouse), including fleas associated with prairie dogs, which we attributed to their frequent use of burrows. Our results support the contention that, at high population densities observed in colonies, grasshopper mice facilitate plague spread by transporting infected fleas between burrows across prairie dog social boundaries represented by coterie territories.

Lewis, R. E., & Eckerlin, R. P. (2013). A Review of the Mustelid-Infesting Fleas in the Family Ceratophyllidae (Insecta: Siphonaptera). *Annals of Carnegie Museum*, 81(2), 119-134.

The three ceratophyllid taxa known to contain specific ectoparasites of mustelid hosts of the genera *Mustela* Linnaeus, 1758, and *Martes* Pinel, 1792, are discussed. The subgenera *Kueichenlipsylla* Smit, 1983, *Paracallosylla* Ioff, 1936, and *Rosickyiana* Smit, 1972, are elevated to generic status. Individual accounts include generic diagnoses, more detailed descriptions of both sexes, where available, and illustrations of diagnostic characters employed in separating the genera. A key is provided that separates both the genera and the species involved. These are: *Kueichenlipsylla atrox* (Jordan, 1925), known only from the Canadian Sub-Region of the Nearctic Region; *Paracallosylla kaznakovi* (Wagner, 1928), restricted to the higher elevations of Central Asia; and *Rosickyiana lunata* (Jordan and Rothschild, 1920), known from a number of collections from western Europe, east to Alaska and central Canada. The status of these taxa is reviewed and their modified nomenclatural ranking proposed. New synonymy is proposed for *Ceratophyllus tundrensis* Holland, 1945, as a junior synonym of *Rosickyiana lunata* (Jordan and Rothschild, 1920).

S.G. MEDVEDEV. 2013. PALAEARCTIC CENTRES OF THE TAXONOMIC DIVERSITY OF FLEAS (SIPHONAPTERA). Entomologicheskoe obozrenie. T. XCII, N 4, P. 684-702.

The Palaearctic flea fauna comprises 921 species and 594 subspecies from 96 genera of 10 families. 858 species (94 %) from 43 genera are Palaearctic endemics and comprise 40 % of Palaearctic Hystrihopsyllidae, 24 % of Ceratophyllidae, and 20 % Leptopsyllidae. Ranges of 581 species (63 % of the Palaearctic fauna) are situated within one province or a subregion of the Palaearctic.

Species with ranges including a part of Asia (592) comprise 87 % of the total fauna; 72 % of the species (517) are endemic to the Palaearctic. The largest centres of the taxonomic diversity of Palaearctic fleas are situated in the East Asian, Central Asian, and Turano-Iranian subregions. 320 species of fleas (of them 214 endemic) from 59 genera (8 endemic) are known from the East Asian Subregion; 270 species (over 120 endemic) from 54 genera (5 endemic) are distributed in the Central Asian Subregion. The Turano-Iranian fauna comprises 213 species (103 endemic) from 47 genera (3 endemic); in the closes to the Russian borders Turanian Subprovince, about 160 species occur, one-third of which (52 species, or 33 %) are endemic. Additional 69 species are endemic to the entire Asian part of the Palaearctic. Extra-Asian and extra-Siberian ranges are known for 190 flea species. In the western Palaearctic, 76 species are endemic to the European Province, and 57 species, to the Mediterranean; 36 species have Euro-Mediterranean distribution. The fauna of the Saharo-Arabian Subregion comprises 30 species (12 are endemic to it), 6 species have ranges of the Mediterranean—Saharo-Arabian type.

Scenarios of the origin of the Siphonaptera at the Triassic/Jurassic boundary are hypothesized. Formation of the Palaearctic flea fauna was mostly supported by the Asian—Indo-Malayan and East Asian—Western American palaeofaunal centers of the flea taxonomic diversity. The long history of the faunal exchanges between eastern Palaearctic and western Nearctic is witnessed by the 701 distribution of the parasites of the rodents and insectivorous mammals, fleas of the genera *Stenoponia*, *Rhadinopsylla*, *Nearctopsylla* and *Catallagia*, belonging to several subfamilies of the Hystrihopsyllidae,

as well as members of a number of other flea families.

A great number of endemic species in the genera *Palaeopsylla* and *Ctenophthalmus* (Hystrichopsyllidae) both in the European and Asian parts of the Palaearctic can be explained by the junction of the European and Asian continental platforms in the late Cretaceous and their subsequent isolation during the Paleocene. A considerable contribution to the flea fauna in the Russian territory was made by the East Asian—Nearctic centre of the taxonomic diversity, with a smaller role of the European palaeofauna. Immigration of species of the family Pulicidae from the Afrotropical Region is restricted to southern territories of Russia.

S.G. Medvedev, and B.K. Kotti. 2012. Patterns of Formation of the Flea (Siphonaptera) Fauna in the Caucasus.

Parazitologiya Vol. 45, No. 6, pp. 470–487. (Russian)

Entomological Review 92(4), 409-421.

Fleas of the Caucasus belong to 155 species of 40 genera, constituting 17% and 43% of the species and generic composition of the Palaearctic fauna, respectively. The Caucasian fauna includes 23 endemic species but no endemic genera or subgenera. In the number of species, the Caucasian fauna is similar to that of the Mediterranean Subregion and is significantly poorer than the faunas of the Euro-Siberian (by 2.2 times) and Irano-Turanian (by 1.7 times) Subregions. Based on taxonomic diversity, we can propose a hypothesis on the West and East Palaearctic sources of the Caucasian fauna. The West Palaearctic source has determined the distribution of pulicomorph fleas of the families Pulicidae and Coptopsyllidae from Africa, on the one hand, and of fleas of the genera *Ctenophthalmus* and *Palaeopsylla* from Europe, on the other hand. Fleas of the Holarctic genera, such as *Ceratophyllus* and *Megabothris*, entered the Caucasus by the north Asian route; fleas of the genera *Neopsylla*, *Rhadinopsylla*, and *Hystrichopsylla* migrated to the Caucasus from east and central Asia by the south Asian route, through Middle and Western Asia.

Medvedev, S. G. and B. K. Kotti. 2013. Host Associations and Origin in the Formation of the Caucasian Fauna of Fleas (Siphonaptera). Entomological Review 93 (3): 293-308.

Medvedev, S. G. 2013. Taxonomic Composition and Zoogeographic Characteristics of the Flea Fauna (Siphonaptera) of Russia. Entomological Review 93 (5): 595-607.

Mehlhorn, H. (2012). Fleas as Underestimated Vectors of Agents of Diseases. (pp. 301-328), in Arthropods as Vectors of Emerging Diseases. Springer Berlin Heidelberg.

Fleas occur worldwide and specimens of their 2,500 species suck blood daily several times at practically all warm-blooded hosts. Therefore, they may become vectors of practically all agents of diseases that are bound to the blood of this armada of varying blood donors. Since the spectacular and very controversial discovery in the year 1898 that the tropical rat flea is the vector of the plague bacterium, many other bacteria and

rickettsial species were found in fleas. Due to the availability of antibiotics, the threat of flea bites had considerably decreased and the dangers of fleas were neglected even for many decades. However, since the resistance of bacteria against antibiotics increases constantly and since it was shown in transmission experiments that also viruses can easily be transmitted by fleas and their feces, the status of flea bites must be considered as being much more dangerous as before. This chapter summarizes a long list of agents of diseases that lurk in fleas, and thus it becomes clear that flea control has become an important task, especially in times of intense globalization.

Erica L. Mize, Jean I. Tsao, and Brian A. Maurer. 2011. Habitat correlates with the spatial distribution of ectoparasites on *Peromyscus leucopus* in southern Michigan. *Journal of Vector Ecology* 36 (2): 308-320.

The goal of this study was to evaluate the role of habitat in determining ectoparasite distribution of *Peromyscus leucopus*. We tested the hypothesis that ectoparasite occurrence is associated with particular host environments and this association is stronger for ectoparasites with limited interactions (i.e., ticks) than those with frequent interactions (i.e., lice).

Ectoparasites from three different groups (Acari, Siphonaptera, and Phthiraptera) were collected from *P. leucopus* inhabiting a number of forested habitats in southern Michigan. Measurements of plant species structure and composition were collected and models were developed using quadratic discriminant function analysis to determine if habitats associated with ectoparasite presence were different from those associated with their absence. Mice parasitized by ticks were more likely to be found in areas having undergone a recent disturbance. Mice parasitized by ticks, fleas, and lice were more likely to be found in areas having tree species associated with dry soils. Our results show there is a distinct difference in habitats associated with the presence of ectoparasites, though we did not observe a stronger association of host habitat for ticks than for fleas or lice. This implies habitat should be included as an important component of assessments of the spatial distribution of ectoparasites.

Ratnaweera, P.B. M.R. Wijesinghe & P.V. Udagama-Randeniya (2010). Parasitic associations of a threatened Sri Lankan rainforest rodent, *Mus mayori pococki* (Rodentia: Muridae). *Journal of Threatened Taxa* 2(6): 901-907.

Parasitic investigations of the subspecies *Mus mayori pococki* were carried out in four rainforests that included two man and biosphere reserves and two forest reserves of Sri Lanka from October 2006 to August 2007. Rodents were live-trapped using Shermans [Sherman] traps.

Of the 117 individuals of *M. m. pococki* captured, 73% were infested with four types of ectoparasites mites of the genus *Echinolaelaps*, a louse *Polyplax serrata*, a larval stage of hard tick *Ixodes* and pseudoscorpions of the genus *Megachernes*. [see Editor's note below.] Mites were the most abundant ectoparasite of this rodent host.

Faecal examination revealed the presence of a nematode larva of the Order

Strongylida and five types of parasitic ova; three nematode ova types i.e. strongyle, strongyloides, ascarid types, and cestode and mite ova. In comparison to the non-infested hosts, those infested did not show a significant difference in body weight and size. Both sexes had an equal probability of being exposed to ectoparasites. The present study on the parasitic investigations of *M. m. pococki* reports four new host-ectoparasite and six new endoparasitic records for the Sri Lankan rodent host.

[Editor's note: Were the pseudoscorpions commensalists or ectoparasites? No fleas seem to have been found.]

Romain, K. S., Tripp, D. W., Salkeld, D. J., & Antolin, M. F. (2013). Duration of Plague (*Yersinia pestis*) Outbreaks in Black-Tailed Prairie Dog (*Cynomys ludovicianus*) Colonies of Northern Colorado. *EcoHealth*, 1-5.

Plague, caused by the bacterium *Yersinia pestis*, triggers die-offs in colonies of black-tailed prairie dogs (*Cynomys ludovicianus*), but the time-frame of plague activity is not well understood. We document plague activity in fleas from prairie dogs and their burrows on three prairie dog colonies that suffered die-offs. We demonstrate that *Y. pestis* transmission occurs over periods from several months to over a year in prairie dog populations before observed die-offs.

Sanchez J.P & M. Lareschi. 2013. The fleas (Insecta: Siphonaptera) parasites of sigmodontine rodents (Cricetidae) from Northern Patagonia, Argentina. *Comparative Parasitology* 80 (1) (110-117).

Bazán-León E., M. Lareschi, J. Sanchez, G. Soto-Nilo, I. Lazzoni, C. Venegas and R. A. Vásquez. 2013. Assemblage of fleas (Insecta: Siphonaptera) associated with non-flying small mammals communities from northern and central Chile; with new host and locality records. *Medical and Veterinary Entomology* 27 (3): 450-459 DOI: 10.1111/mve.12005

López Berrizbeitia MF, Lareschi M, Sánchez RT & Díaz MM. 2013. Los ectoparásitos de los roedores sigmodontinos (Cricetidae) de La Rioja: resultados preliminares. *Revista Argentina de Parasitología* 3: 40-44.

Colombo V.C., Lareschi, M., Monje L.D., Nava S., Antoniazzi L.R., Beldomenico, P.M. & A.A. Guglielmone. 2013. Garrapatas (Ixodida) y ácaros (Mesostigmata) parásitos de roedores sigmodontinos del Delta del Paraná, Argentina. *RIA*. In press.

López Berrizbeitia, MF, MM Diaz, RM Barquez & M. Lareschi. 2013. Pulgas (Siphonaptera) parásitos de roedores (Rodentia: Cricetidae) de la provincia de Salta, Argentina: nuevos registros de distribución. *Revista de la Sociedad Entomológica Argentina*. In press.

Pucu E, M. Lareschi & SL Gardner. 2013. Bolivian Ectoparasites: A survey of the fleas

of *Ctenomys* (Rodentia: Ctenomyidae). Comparative Parasitology. In press.

Sanchez J.P & M. Lareschi. 2014. New records of fleas (Siphonaptera: Ctenophthalmidae, Rhopalopsyllidae, Stephanocircidae) from Argentinean Patagonia with remarks on the morphology of *Agastopsylla boxi* Jordan and Rothschild, 1923 and *Tiarapsylla argentina* Jordan, 1942. Revista Mexicana de Biodiversidad. In press.

Sikes DS, Stockbridge J (2013) Description of *Caurinus tlagu*, new species, from Prince of Wales Island, Alaska (Mecoptera, Boreidae, Caurininae). ZooKeys 316: 35–53. doi: 10.3897/zookeys.316.5400

A new species of the cryptic, minute, wingless, and enigmatic taxon *Caurinus*, and the second for the subfamily Caurininae, is described from Prince of Wales Island in the Alexander Archipelago, Alaska. It is distinguished from its only congener, *Caurinus dectes* Russell, 1979b, which occurs 1,059 km southeast in Oregon and Washington, based on external morphology and sequences of the mitochondrial gene cytochrome oxidase II. These two species are probably evolutionary relicts – the only known members of a clade dating to the Late Jurassic or older.

[Editor's extended extract: "Russell (1979a,b, 1982) described the monotypic subfamily Caurininae, genus and species *Caurinus dectes*, known only from Oregon and Washington, and later described by Beutel et al. (2008) as “arguably one of the most bizarre and cryptic species of Mecoptera and endopterygote insects.” Indeed, members of the genus do not key to any order in most keys to insect orders because they lack a produced rostrum, typical of the order Mecoptera, and lack the diagnostic traits that would place them within *any* insect order containing flightless adults with rudimentary or vestigial wings. However, they do share with members of the family Boreidae a very distinctive wing morphology and sexual dimorphism in which the adult females are nearly wingless while the males bear shortened scissor-like wings, useless for flight, that bear spines for grasping females during mating. The placement of *Caurinus* within the Mecopteran family Boreidae as the sister taxon to the Boreinae (*Boreus* 26 spp., *Hesperoboreus* 2 spp. [Penny 2013]), is apparently well established based on morphological study (Russell 1979a, b, Beutel et al. 2008, Friedrich et al. 2013) and molecular phylogenetics (Whiting 2002).

However, despite recent efforts, the genus remains enigmatic due to its preponderance of plesiomorphic and autapomorphic traits (Beutel et al. 2008). The close relationship of the Mecoptera with the fleas, order Siphonaptera, is of particular evolutionary interest (Grimaldi and Engel 2005, Whiting 2002, Trautwein et al. 2012)....

This suggests the split between the genus *Caurinus* and the remaining boreids likely predates the oldest confirmed boreid fossil, *Palaeoboreus zherichini* Sukatsheva & Rasnitsyn, of the Late Jurassic (Grimaldi and Engel 2005) which appears to be a boreine due to its size and external ovipositor, although it lacks the produced rostrum typical of extant species (Russell pers. com.). If confirmed, such a great age (>145 Ma) for a genus of two extant species would make the lineage an evolutionary relict and its species

certainly deserving of conservation attention (Habel and Assmann 2010, Naskrecki 2011)."]

Russell, L. K., R. Dallai, M. Gottardo, and R. G. Beutel. 2013. The Sperm Ultrastructure of *Caurinus dectes* Russell (Mecoptera: Boreidae) and its Phylogenetic Implications. *Tissue and Cell* 45 (6): 397-401.

Loren Russell and Derek Sikes. 2013. How to collect *Caurinus* (Mecoptera, Boreidae), <http://vimeo.com/68819818#at=0> (accessed 4 January 2014).

Video of collecting *Caurinus* on Prince of Wales Island, Alaska, showing the conifer forest habitat for boreal scorpionflies (Boreidae) on the leafy liverwort *Scapania bolanderi*. (~~frame of video below~~) [picture may be copyright protected so it is removed by editor. Please see internet site for original video.]

Votýpka, J., Suková, E., Kraeva, N., Ishemgulova, A., Duží, I., Lukeš, J., & Yurchenko, V. (2013). Diversity of Trypanosomatids (Kinetoplastea: Trypanosomatidae) Parasitizing Fleas (Insecta: Siphonaptera) and Description of a New Genus *Blechomonas* gen. n. *Protist*, 164(6), 763-781.

To further investigate the diversity of Trypanosomatidae we have examined the species present within the flea (Siphonaptera) population in the Czech Republic. Out of 1549 fleas, 239 were found to be infected by trypanosomatids. Axenic cultures were established from 90 infected specimens and 29 of them were further characterized. Molecular phylogenetic analysis of the SL RNA, gGAPDH, and SSU rRNA genes revealed a striking diversity within this group and analyzed isolates were classified into

16 Typing units (TUs) of which 15 typified new species. In addition to one *Trypanosoma* species, two TUs grouped within the sub-family Leishmaniinae, two clustered together with *Herpetomonas*, whereas 11 TUs formed a novel clade branching off between *Trypanosoma* spp. and remaining trypanosomatids. We propose to recognize this clade as a new genus *Blechomonas* and a new subfamily Blechomonadinae, and provide molecular and morphological description of 11 TUs representing this genus. Our finding of such an ancient host-specific group sheds new light at the origin of Trypanosomatidae and the roots of dioxenous parasitism. The strict host restriction of *Blechomonas* to Siphonaptera with adult fleas' dependence on blood meal may reflect passing of parasites from larvae through pupae to adults and implies potential transmission to the warm-blooded vertebrates.

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Fleas in Art and History

"The mammoth, an extinct big cousin of the elephant, responded to the mosaic of wetlands and ridges, whereas its flea responded to individual hairs and varying hair densities on the mammoth's knee. The two animals responded to different grain sizes. Indeed, the mammoth went extinct, but the fate of the mammoth flea is still unknown."

Forman, R. T. (1995). Land mosaics: the ecology of landscapes and regions. Cambridge University Press. (pg. 10).

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Tom Sawyer Abroad, Chapter VII: TOM RESPECTS THE FLEA
by Mark Twain

http://www.mtwain.com/Tom_Sawyer_Abroad/6.html

"NOON!" says Tom, and so it was. His shadder was just a blot around his feet. We looked, and the Grinnage clock was so close to twelve the difference didn't amount to nothing. So Tom said London was right north of us or right south of us, one or t'other, and he reckoned by the weather and the sand and the camels it was north; and a good many miles north, too; as many as from New York to the city of Mexico, he guessed.

Jim said he reckoned a balloon was a good deal the fastest thing in the world, unless it might be some kinds of birds -- a wild pigeon, maybe, or a railroad.

But Tom said he had read about railroads in England going nearly a hundred miles an hour for a little ways, and there never was a bird in the world that could do that -- except one, and that was a flea.

"A flea? Why, Mars Tom, in de fust place he ain't a bird, strickly speakin' --"

"He ain't a bird, eh? Well, then, what is he?"

"I don't rightly know, Mars Tom, but I speck he's only jist a' animal. No, I reckon dat won't do, nuther, he ain't big enough for a' animal. He mus' be a bug. Yassir, dat's what he is, he's a bug."

"I bet he ain't, but let it go. What's your second place?"

"Well, in de second place, birds is creturs dat goes a long ways, but a flea don't."

"He don't, don't he? Come, now, what IS a long distance, if you know?"

"Why, it's miles, and lots of 'em -- anybody knows dat."

"Can't a man walk miles?"

"Yassir, he kin."

"As many as a railroad?"

"Yassir, if you give him time."

"Can't a flea?"

"Well -- I s'pose so -- ef you gives him heaps of time."

"Now you begin to see, don't you, that DISTANCE ain't the thing to judge by, at all; it's the time it takes to go the distance IN that COUNTS, ain't it?"

"Well, hit do look sorter so, but I wouldn't 'a' b'lieved it, Mars Tom."

"It's a matter of PROPORTION, that's what it is; and when you come to gauge a thing's speed by its size, where's your bird and your man and your railroad, alongside of a flea? The fastest man can't run more than about ten miles in an hour -- not much over ten thousand times his own length. But all the books says any common ordinary third-class flea can jump a hundred and fifty times his own length; yes, and he can make five jumps a second too -- seven hundred and fifty times his own length, in one little second -- for he don't fool away any time stopping and starting -- he does them both at the same time; you'll see, if you try to put your finger on him. Now that's a common, ordinary, third-class flea's gait; but you take an Eye-talian FIRST-class, that's been the pet of the nobility all his life, and hasn't ever knowed what want or sickness or exposure was, and he can jump more than three hundred times his own length, and keep it up all day, five such jumps every second, which is fifteen hundred times his own length. Well, suppose a man could go fifteen hundred times his own length in a second -- say, a mile and a half. It's ninety miles a minute; it's considerable more than five thousand miles an hour. Where's

your man NOW? -- yes, and your bird, and your railroad, and your balloon? Laws, they don't amount to shucks 'longside of a flea. A flea is just a comet b'iled down small."

Jim was a good deal astonished, and so was I. Jim said: "Is dem figgers jist edjackly true, en no jokin' en no lies, Mars Tom?"

"Yes, they are; they're perfectly true."

"Well, den, honey, a body's got to respec' a flea. I ain't had no respec' for um befo', sca'sely, but dey ain't no gittin' roun' it, dey do deserve it, dat's certain."

"Well, I bet they do. They've got ever so much more sense, and brains, and brightness, in proportion to their size, than any other cretur in the world. A person can learn them 'most anything; and they learn it quicker than any other cretur, too. They've been learnt to haul little carriages in harness, and go this way and that way and t'other way according to their orders; yes, and to march and drill like soldiers, doing it as exact, according to orders, as soldiers does it. They've been learnt to do all sorts of hard and troublesome things. S'pose you could cultivate a flea up to the size of a man, and keep his natural smartness a-growing and a-growing right along up, bigger and bigger, and keener and keener, in the same proportion -- where'd the human race be, do you reckon? That flea would be President of the United States, and you couldn't any more prevent it than you can prevent lightning."

"My lan', Mars Tom, I never knowed dey was so much TO de beas'. No, sir, I never had no idea of it, and dat's de fac'."

"There's more to him, by a long sight, than there is to any other cretur, man or beast, in proportion to size. He's the interestingest of them all. People have so much to say about an ant's strength, and an elephant's, and a locomotive's. Shucks, they don't begin with a flea. He can lift two or three hundred times his own weight. And none of them can come anywhere near it. And, moreover, he has got notions of his own, and is very particular, and you can't fool him; his instinct, or his judgment, or whatever it is, is perfectly sound and clear, and don't ever make a mistake. People think all humans are alike to a flea. It ain't so. There's folks that he won't go near, hungry or not hungry, and I'm one of them. I've never had one of them on me in my life."

"Mars Tom!"

"It's so; I ain't joking."

"Well, sah, I hain't ever heard de likes o' dat befo'." Jim couldn't believe it, and I couldn't; so we had to drop down to the sand and git a supply and see. Tom was right. They went for me and Jim by the thousand, but not a one of them lit on Tom. There warn't no explaining it, but there it was and there warn't no getting around it. He said it

had always been just so, and he'd just as soon be where there was a million of them as not; they'd never touch him nor bother him.

We went up to the cold weather to freeze 'em out, and stayed a little spell, and then come back to the comfortable weather and went lazying along twenty or twenty-five miles an hour, the way we'd been doing for the last few hours. The reason was, that the longer we was in that solemn, peaceful desert, the more the hurry and fuss got kind of soothed down in us, and the more happier and contented and satisfied we got to feeling, and the more we got to liking the desert, and then loving it."

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In this short story by the author of "The Moonstone" and "Woman in White", the police use fleas as a search warrant:

" "I swear to you, Monsieur le Sous-préfet [of police], he is not here! He – "

"I swear to you, Monsieur le Garçon, he is. He slept here – he didn't find your bed comfortable – he came to us to complain of it – he is here among my men – and here am I ready to look for a flea or two in his bedstead." "

- "A Terribly Strange Bed", by William Wilkie Collins (1824 – 1889).

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"The nuns arrived at Matanzas Bay on January 1, 1899, although they remained on ship until January 6, riding in an army ambulance to Mass in a Spanish hospital each day. Once on land, they assumed charge of a government hospital where they continued their work of mercy. In addition, they opened up a diet kitchen for the patients. Now, however, the nuns encountered intense heat during the day as opposed to the extreme cold they had endured in Kentucky. There was also a problem with an unwelcome invader: "The nights are cool," sister observed, "but the fleas have us about devoured." ...

But Civil War nuns had their problem with fleas too as reported by Sr. Camilla O'Keefe, Daughter of Charity, when nursing at Gettysburg: "The sisters also brought plenty of the vermin along in their clothes. I shudder to think of this part of the sister's sufferings while they were serving in the military hospitals, especially in the field tents in Gettysburg;" "

- *Mercedes Graf*. 2002. Band of Angels, Sister Nurses in the Spanish-American War, Part 1, 2. [Prologue Magazine](http://www.archives.gov/publications/prologue/2002/fall/band-of-angels-1.html#f56) Fall, Vol. 34, No. 3 [unpaginated]. U.S. National Archives and Records Administration, College Park, MD, USA. <http://www.archives.gov/publications/prologue/2002/fall/band-of-angels-1.html#f56>

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The Ehrenfest Fleas

(P. Ehrenfest and T. Ehrenfest. 1907. *Phys. Z.* 8, 311.)

Ambegaokar, V., & Clerk, A. A. 1999. Entropy and time. *American Journal of Physics*, 67, 1068.

"... consider a system consisting of a subsystem of 50 fleas whose "states" are residence on dog A or dog B, which we shall call Anik and Burnside, sleeping side by side. To simulate molecular agitation, we suppose that the fleas jump back and forth between the dogs.

Now we need something that plays the role of the rest of the system. Let us suppose that the fleas are each equipped with a number, and have been trained to jump when their number is called. The "environment" agitating the fleas, which is like a heat reservoir, will be something that calls out numbers at random, and our closed system will be the fleas and the reservoir. To simulate the approach to equilibrium, it is necessary to start from a configuration that almost never occurs in the maximally random state of affairs.

Suppose that in the beginning Anik has no fleas at all. We agitate the fleas by having a computer generate random numbers between 1 and 50 and transferring the flea with this number from one dog to the other. At every step we record the total number (between 0 and 50) of fleas on Anik, but not their labels. This way has the practical advantage that we do not have to keep track of the 250 ways of assigning 50 labelled fleas to the two dogs. It also means that we are following only the "macrostate." "

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"At a fundamental level, the mission of Naess's ecosophy is to expand the sphere of objects with which people identify. He believes that 'identification elicits intense empathy' and that humans remain indifferent to that which they take to be utterly different than themselves (Naess, 1995: 15). To support this position, Naess shares a personal anecdote about a flea that suddenly landed in a sample of acid chemicals, which Naess was studying under a microscope. He claims, 'If I was alienated from the flea, not seeing intuitively anything even resembling myself, the [flea's] death struggle would have left me indifferent' (Naess, 1995: 15). This anecdote, a vital illustration of Naess's thought, brings us to the most important difference between his ecosophy and the ecosophy of Felix Guattari. Naess calls for an expansion of the self via identification ("Self-realisation"), whereas Guattari (and Deleuze) valorise autopoietic processes that perform a dissolution of the self via disjunction ("becoming-other"). In other words—in a Guattarian reworking of the flea anecdote—I would not look for elements of the flea that remind me of myself; rather, I would receive the flea in its alterity and encounter aspects of the fleas that are completely different from myself, so as to "become-flea": to

introduce the flea's manner of existence into the way I think and live. [3] Initially, the difference between Naess's identification and Guattari's autopoiesis may seem trivial. This minor difference, however, actually lays out two divergent, even conflicting, paths for diagramming the production of subjectivity."

- Tinnell, J. 2011. Transversalising the Ecological Turn: Four Components of Felix Guattari's Ecosophical Perspective. The Fibreculture Journal FCJ-121 issue 18: 35-64. issn: 1449 1443. <http://eighteen.fibreculturejournal.org/2011/10/09/fcj-121-transversalising-the-ecological-turn-four-components-of-felix-guattari-%e2%80%99s-ecosophical-perspective/>

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(on the next page) "The Flea Catcher" (c.1655), by Gerard ter Borch (1617-1681, Netherlands), 27.1 x 34.4 cm, oil on canvas.



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Flea research of historical interest

LYON, HAROLD. 1915. NOTES ON THE CAT FLEA (*CTENOCEPHALUS* [= *Ctenocephalides*] *FELIS* BOUCHÉ]). *Psyche*, pg. 124-132.

[Editor's note: the following extract describing the larva of the cat flea has various typographical errors, such as cited names. Please check the original for accuracy. Lyon's account is interesting not only for the morphological details, but as well the description of how the flea larva uses its parts to move and eat.]

"The broad convex upper lip or labrum of the mouth of the cat flea [larva] is slightly emarginated in the middle with two small setae at the center of either lobe ... There are also two small hairs on either side. The labrum is used extensively in locomotion and in connection with the rest of the head to lift and turn over particles of food and in burrowing. In locomotion the two small setae or bristles are fastened into the surface upon which the larva may be, and act as a brace while the "caudal stylets" at the extremity of the body are being placed in preparation for exerting the power to force the body forward. The labrum is then relaxed and the larva darts forward in a rapid wriggling motion. On the underside of the labrum is a curved row of bristles which partially line the upper surface of the mouth and when closed meet the bunches of bristles on the ends of the maxillae, thus holding the food in place in front of the mandibles.

The biting mandibles ... which are situated well inside the mouth, are strongly chitinized, broad and triangular and seem to be very powerful for such a larva, appearing more like those of Coleoptera than Diptera. The outer small end tapers gradually back and ends in a broad base with a single large lobe. The teeth appear, when the mandibles are in normal position, to be situated, not on the inside edge but somewhat back and on opposite sides of each mandible. Thus, when the two mandibles are closed, they overlap each other. The teeth, six in number, are blunt and not sharp, as Kunckel states, nor do the mandibles end in a sharp point. The lobe at the base of the mandibles serves for the attachment of muscles and as a pivot by which they are moved.

Kunckel describes the maxillae (Fig. B and C) as being sectors of a circular saw with the teeth on the greater part of their outline; Taschenberg, as being sickle-shaped on the outer edge, circular in shape on the inner edge which is provided with numerous fine teeth. The two maxillae are provided on their inner cutting surfaces with five sharp-like saw teeth which point upwards. On the ends of the maxillae are clumps of fine bristles or "brushes." These work against the curved line of hairs on the roof of the mouth.

The bristles on the outer surface of the maxillae interlock when in normal position. In the basal portion of the maxillae, is a part, more heavily chitinized than the rest, which tapers up into a point on either side. Perhaps this is the sickle-shaped affair of Taschenberg.

This is evidently a brace for the maxillae, providing a suitable support for the attachment of muscles at the base and for the support of the palpi, which are two-jointed. The basal joint is short, broad and higher at one end than at the other. The second

joint is short and stout, enlarging at its tip an oblique, flat end, bearing five small papilliform processes of unequal length which radiate somewhat.

The labium is situated well under the head (Fig D). Laboulbene says that it is boat-shaped and divided in front, with two palpiform lobes. Kunckel says that it is tongue-shaped with two very fine points making it bifid, having at its base two humps provided with two long and two short bristles. The labium is very small, narrow, slightly swelling near the end, with a blunt point which is turned upwards. Its bifid appearance is due to the presence of the labial palpi which are situated just back of the tip on the slightly enlarged portion and not at the base of the labium. The palpi consist of a short thick basal joint with four stout bristles, two of which are short and appear almost together, while the other two are longer and spreading. Like the terminal bristles of the antennae, they do not taper. The base of the labium appears to end in the suture which extends medially the full length of the under side of the head."

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Obituaries

Robert “Bob” Stephen George (1921-2013)
UK Flea Recorder

By A. Whitaker

Few people are gifted enough to have either an amazing career or a lifelong passion, but Bob George was one of those extraordinary people who had both. Born on 5th May 1921 in Gloucester to an architect father, the young Robert Stephen George showed early promise as a budding biologist with his interest in collecting all things

natural – mosses, spiders, birds' eggs, butterflies, fossils, newts, etc. As a sixth former, he recorded over 500 species of flowering plants, data that was later used in the publication *Flora of Gloucestershire*.

On leaving school, war broke out, and in March 1941, he joined the RAF, training as a pilot in the USA. On his return to the UK, he took to flying Spitfires, and was based with the No.616 South Yorkshire Squadron, attacking road and rail communications in Northern France. In 1944, he converted to the Meteor [one of the few jets of World War 2], and moved to Kent to intercept bombs. Towards the end of the war he was carrying out ground attack missions in Germany and, as an experienced fighter pilot, he started instructing new pilots, before leaving the RAF in 1946.



In 1943, he had married Dorothy Benoist, and perhaps favouring a quieter life after the war, decided to pursue a career in education, teaching biology at his former school, and later becoming Principal of Adult Education at a college in Cambridgeshire. By 1949, he was already rekindling his fascination with natural history, but was inclined towards beetles.

However, this all changed when he lived in a flat inconveniently located on a “mouse-motorway” between a restaurant and a butcher’s shop. Over three nights, and with three traps, he caught 35 mice, one of which had 15 fleas on it. He sent the fleas to Frans Smit, curator of the Rothschild collection of Fleas at Tring Museum [and founder of *Flea News*] who identified them for him, and told him that two species were new to Gloucestershire. A few days later he collected a few hundred fleas from the nest of a long-tailed tit located in a coal heap in the playground where he taught, and this haul led to another two new species recordings. So, within the space of a few days, Bob’s collecting had resulted in four new flea species records for Gloucestershire!

Encouraged by Smit and Miriam Rothschild, another expert on fleas [*Flea News* 62, June 2008], Bob was hooked, realising that by studying fleas he could make a real

difference to the recording of British fauna, whereas the showy beetles were already heavily oversubscribed with enthusiastic amateurs. His first paper on fleas was *The Siphonaptera of Gloucestershire*, and he went on to publish over 50 papers on fleas, including the first *Provisional Atlas of the Distribution of British Fleas* in 1974 with the Biological Records Centre at Monks Wood. By 2006, he had identified and recorded several hundreds of thousands of specimens, and a second *Atlas of the Fleas (Siphonaptera) of Britain and Ireland* was published in 2008, which he felt reflected much better the distribution of fleas, rather than the distribution of collectors of fleas. The second *Atlas* also coincided with the second *Handbook of British Fleas* published in 2007 by the Royal Entomological Society and the Field Studies Council, the first Handbook having been published by Smit in 1957. As the author of this Handbook, this is where my connection with Bob began.



As soon as he heard of my appointment through *Flea News*, he immediately wrote to me, expressing his delight that Smit's original handbook was to be updated. Thereafter, we regularly communicated, and I received many letters written in neat black handwriting on pink paper (complete with the headline "Recording Britain's Fleas" and a line drawing of a female cat flea). Bob was immensely helpful in reading and commenting on the draft manuscript, and in pointing me to references I may have omitted. He also allowed me to use a wonderful poem written and given to him by Smit in 1985. I reciprocated by sending him copies of publications which he did not have, and supplying the front cover photograph for his updated *Atlas*.

Although something of a local celebrity in Bournemouth, and regularly featured in the local papers, his wife did not always share his passion. She was horrified one day when she overheard some people passing the house, one of them commenting, "That's

the house which contains thousands of fleas”! They were indeed correct, and Bob’s initial 15 fleas found on a single mouse pales in comparison to the 5,080 fleas found in a single house martin’s nest. This figure was then trumped by the 7,116 fleas collected from a single hedgehog sent by a hedgehog sanctuary in Guernsey – for the next year or so they sent him the fleas from every hedgehog they received, 30 vials at a time. Of the 1000’s received, all but 10 were identified by him as the hedgehog flea [*Archaeopsylla erinacei erinacei* (Bouché, 1835)]. He also debunked the generally held belief that all hedgehogs are loaded with fleas, apparently, a quarter of the hedgehog population had 100-800 fleas, but a third had no fleas at all.

Over the years as the UK Flea Recorder, Bob had an army of collectors working around the country, mainly bird and bat enthusiasts, who would regularly send him fleas, often alive and still attached to parts of their host or host’s nest, much to the disapproval of the post office. His wife and children were not too pleased either when, during a week that he was away from home, the inhabitants of a badly packed bird’s nest escaped.

Fleas are a much-understudied order of insect, but most British entomologists will no doubt have had a connection with Bob through Marris House Nets, the company he established in the 1970’s specialising in the manufacture and supply of entomological nets. Through a local contact in Bournemouth, Bob had become interested in Mymarids [Mymaridae], a family of Chalcid [superfamily Chalcidoidea] wasps, commonly called “fairyflies” or “fairy wasps”. On visiting the Natural History Museum’s wasp experts in order to learn the best way of collecting these tiny flying insects, he was shown a Malaise trap, and told that they had to be hand-made in-house and then dyed black. Bob was convinced that he could do better, and within a week had found a supplier of the necessary black material, together with someone to make the nets. However, it was only possible to buy the material in bulk, so he was persuaded to commit himself to making and selling a large quantity of nets. Thus, Marris House Nets was conceived, and Malaise traps are still the most commonly used trap for sampling airborne insects. He even used Mymarids as the inspiration for its logo – the image of a naked female fairy with flowing locks and four wings against a hatched triangle will be very familiar to most UK entomologists!

Bob was a Fellow of the Linnean Society, a Fellow of the Royal Entomological Society, and a founding member of the Gloucester Wildlife Trust. He was awarded Honorary Membership of the National Biodiversity Network Trust in 2006.

Bob’s wife Dorothy died in May 2005, and he is survived by a son and two daughters, plus three grandchildren.

[R. S. George, and P. T. Harding. 2008. Atlas of the Fleas (Siphonaptera) of Britain and Ireland. Field Studies Council Publ., Biological Records Centre (Centre for Ecology and Hydrology), Shrewsbury, United Kingdom. ISBN 978-0-95576-723-4, 98pp, £8.00.]

[Editor's note: For citation information, see the British Journal of Entomology and Natural History that will publish the obituary. Bob George's Flea Atlas of Britain and Ireland is available at

<http://www.ceh.ac.uk/products/publications/fleaatlsofbritainandireland.html> . The Flea database at UK National Biodiversity Network is at <http://data.nbn.org.uk/> .

It is interesting to note that the UK has a network of biodiversity recorders for many of the island's taxa, including plants, fungi, and animals. It is remarkable that George's efforts in collecting fleas resulted in forming the UK Flea Recorder position especially for him. This shows that one person's efforts can make a difference!

The United Kingdom flea fauna is arguably the most comprehensively documented in the world. We find this level of interest and organization in biodiversity observations nowhere else, to my knowledge, although ironically humans have extirpated many species years ago, such as bears, boars, and wolves, from Great Britain.

The fairyflies mentioned that Bob George was interested in, are parasitoids on other insects. Fairyflies are probably some of the smallest insects in the world, hence the concern with nets; when full grown, they may be only 0.21 mm long, which is smaller than many single-celled eukaryotes! (Borror et al., Introduction to the Study of Insects).]

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Bob George on How to Collect and Record Fleas

[From George, R. S. 2003. THE FLEAS (SIPHONAPTERA) RECORDED FROM CLACKMANNANSHIRE. Forth Naturalist and Historian Vol. 26, p. 67-69).]

[with photograph of "*Hystrichopsylla t. talpae* (Curtis). Britain's largest flea – a flea on moles, voles, field mice and shrews."]

"In chronological order of publication we get:

(i) From Waterston, (1909): *Ceratophyllus gallinae* (Schrank), common at Kennetpans (near Kincardine-on-Forth), 8-10. viii. 1908, *M. lugubris*, *P. domesticus*, *P. coeruleus*. A female also among dead leaves at the same place. *C. farreni*, Roths., 9 male 27 female, *C. urbica*, Kennetpans, 10.viii. 1908. *C. hirundinis* (Curtis), male and 9 female, *C. urbica*, Kennetpans, 10.viii. 1908.

(ii) Rothschild, (1952) fills out Waterston's 1909 records from material in the collection of the British Museum (Natural History) – now The Natural History Museum, London – thus: *Ceratophyllus gallinae*, 4 male 10 female, *Passer domesticus*, Kennetpans, 8-10.viii. 1908, J. Waterston; 5 male 6 female, *Motacilla alba yarrelli*, Kennetpans, 6.viii.08, J. Waterston; 8 male < female, *Parus coeruleus* nest, 8-10.viii.08, Kennetpans, J. Waterston.

(iii) Smit, (1957), in his immensely useful paper, lists as having seen Clackmannanshire specimens of *Ceratophyllus gallinae*, *C. rusticus* Wagner, *C.f. farreni*,

C. hirundinis and *C. garei* Rothschild but no data is given.

(iv) Thompson, (1958), repeats Waterston's records.

(v) Mead-Briggs and Page, (1964), record: 1 female *Archaeopsylla e.erinacei* (Bouché), 1 male *Nosopsyllus fasciatus* (Bosc) and 4 male 4 female *Spilopsyllus cuniculi* (Dale) from male rabbit, 11.X.62, Harvieston, nr. Dollar. The first two species are stragglers on rabbit, *erinacei* from hedgehog and *fasciatus* from rats. *Spilopsyllus cuniculi* is specific to rabbits but does occur on hares and cats.

(vi) Mead-Briggs, (1964), repeats the above *S. cuniculi* record.

(vii) Usher, (1968), records collecting from mole nests during the previous five years seven species but, unfortunately, the only data he gives are the 10 km squares of the national grid. He collected: *Hystrihopsylla t.talpae* (Curtis), NS99; *Rhadinopsylla pentacantha* (Rothschild), NS99; *Palaeopsylla kohauti* Dampf, NS99; *Palaeopsylla minor* (Dale), NS99; *Ctenophthalmus n.nobilis* (Rothschild), NS99, NT09; *Ct. n.vulgaris* Smit, NS99, NT09; *Ct. bisoctodentatus heselhausi* Oudemans, NS99, NT09.

Finally, two specimens were sent to me for confirmation of identity. They are: *Rhadinopsylla pentacantha* (Rothschild), 1 from mole nest, nr. Kilbagie, – iii.64, M.B. Usher, NS9391 and *Spilopsyllus cuniculi* (Dale), 1 female from body of *Lepus europaeus*, Aitkenhead Forest Mill, 8.i.64, A.R. Mead-Briggs, NS 940940.

This all adds up to a rather poor list and an immense dearth of collecting and much more must be done before a final list can be published. The following points arise:

a) The scope for more collecting leads the naturalist initially to the common species. For instance no cat fleas have been recorded, they will be found on most cats and dogs and, perhaps, the much less common dog flea. There are two species to be found on shrews, one in sand martin nests, three in house martin nests, two in nesting boxes, up to six in vole nests, one on badgers. Bird nests in damp areas are likely to have up to three species, wheatear nests have another. The nests of Corvidae possibly another couple. Few of our bird fleas and no bat fleas have been recorded from the County.

It must be remembered that bird nests must not be interfered with whilst still in use.

b) Most of the records are old and the species concerned have been recorded from single localities. Re-recording is a necessity.

c) The need for cross-co-operation between naturalists of different disciplines is necessary. The siphonapterist is largely in the hands of mammalogists and ornithologists if his work is to be successful. Members of both groups should become actively interested in the ectoparasites of the vertebrates in which they profess to be involved. Veterinarians and staff at the various kinds of animal refuges could help. Making local lists is an activity in which secondary school pupils can be involved, e.g. my first paper, the first paper on fleas of Gloucestershire, was done with a very considerable amount of assistance of my pupils.

d) The need for authors of papers at all levels to give full data when writing their papers when otherwise much information is lost."

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Publications of K. Hůrka (1931-2004) involving bat parasites

Obituary in HANÁK, V. 2004. In memoriam Professor Karel Hůrka. *Vespertilio* 8: 143–146. ISSN 1213-6123 (excerpt, translated from Czech)

HŮRKA K., 1956: Die Larve des Fledermausflohes *Ischnopsyllus intermedius* (Roth. 1898). *Věst. Čs. Společ. Zool.*, **20**: 372–374. [The larva of the bat flea *Ischnopsyllus intermedius*]

HŮRKA K., 1957: Příspěvek k systematice, faunistice, bionomii a ekologii netopyřích blech v ČSR. *Čs. Parasitol.*, **4**: 145–166. [Contribution to the systematics, fauna, bionomics and ecology of bats fleas in Czechoslovakia]

HŮRKA K., 1958: Hmyzi ektoparasiti netopyřů v oblasti rezervace Velký a Malý Tisý. *Ochr. Přír.*, **13**(4): 96–98. [Insect ectoparasites of bats in Velký a Malý Tisý]

HŮRKA K., 1958: Beitrag zur Nycteribidenfauna Bulgariens (Nycteribiidae, Diptera). *Zool. Listy*, **7**: 231–234. [contribution to Bulgarian Nycteribidenfauna]

HŮRKA K., 1959: *Penicillidia monoceros* Speiser in Czechoslovakia: Contribution to the morphology, bionomy and distribution. *Čas. Čs. Společ. Entomol.*, **56**: 126–128.

HŮRKA K., 1959: Příspěvek k poznání hmyzích ektoparasitů netopyřů Jizerských hor. *Sborn. Severočes. Mus., Přír. Vědy*, **1**: 142–146.

HŮRKA K., 1962: Beitrag zur Nycteribien- und Streblidenfauna Albaniens nebst Bemerkungen zur Fauna von Bulgarien, Ungarn und UdSSR. *Čas. Čs. Společ. Entomol.*, **59**: 156–164.

HŮRKA K., 1963: Bat fleas (Aphaniptera, Ischnopsyllidae) of Czechoslovakia II. Subgenus *Hexactenopsylla* Oud., genus *Rhinolophopsylla* Oud., subgenus *Nycteridopsylla* Oud., subgenus *Dinycteropsylla* Ioff. *Acta Univ. Carol. – Biol.*, **1963**(1): 1–73.

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HŮRKA K., 1963: Ergebnisse der Albanien-Expedition 1961 der Deutschen Entomologischen Institutes. 8. Beitrag. Diptera: Nycteribidae und Streblidae. *Beitr. Entomol.*, **13**: 59–64. 145

HŮRKA K., 1963: Zur Kenntnis der Puparien von *Nycteribia latreilli* (Leach), *N. kolenatii* Theodor, *N. Vexata vexata* Westwood und *N. biarticulata* Hermann nebst Bemerkungen zu den übrigen bekannten Puparien der europäischen Nycteribienarten. *Věst. Čs. Společ. Zool.*, **27**: 46–50. [For knowledge of the puparia of *Nycteribia latreilli* (Leach), *N. kolenatii* Theodore, *N. vexata vexata* Westwood and *N. biarticulata* Hermann, together with comments on the remaining known puparia of European Nycteribienarten.]

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Nixon A. Wilson (1930-2011)
Ectoparasitologist

By R.L. Bossard

Born May 20, 1930, Litchfield, Illinois, USA, son of Benezette and Adrian Barnes Wilson. Married Nell Hines, June 15, 1963, Louisville, Kentucky. Died Dec. 7, 2011, Cedar Valley Hospice Home, Waterloo, Iowa.

Education: A.B. degree, Earlham College, 1952; Master's, Wilderness Management, University of Michigan, 1954; Doctorate, Purdue University, 1961.

Employment: Served in the U.S. Army (Korean War).

Ecologist, Hawaii State Department of Health, Plague Research Unit, 1961 - 1962.

Acarologist, Bernice P. Bishop Museum, Honolulu, 1962 - 1969.

Professor, University of Northern Iowa, 1969 - 1997 (Distinguished Scholar Award, 1991; McKay Award, 1992; Regent's Award for Faculty Excellence, 1993).

Though I never met Dr. Wilson, I exchanged specimens with him. His expertise was remarkable concerning all ectoparasites – fleas, lice, mites, ticks, bugs, beetles, and flies, as well as in the great care he took in collecting. He advised examining individual hosts separately for parasites, recording all ectoparasites on that individual, including the lack of parasites, and avoiding the placing hosts together where one host's parasites could contaminate another host.

Partial Necrology

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Pulex irritans L. was believed to be the only species of *Pulex* in Hawaii until *P. simulans* Baker was discovered in 1963 on a dog at Punaluu, Island of Hawaii. Evidence indicates *P. simulans* has been in Hawaii more than 43 years but was confused with *P. irritans*. In 1964, a survey of *Pulex* infesting dogs on Hawaii showed *P. simulans* widespread, but *P. irritans* present only on the northern half of the island where it occurred most frequently at higher elevation, especially at Kukuihaele. The wider distribution of *P. simulans* included the driest, wettest and warmest areas sampled on Hawaii. At least 3 dogs were infested with both species. *P. simulans* was recorded for the first time from Oahu and Kauai, but not Molokai, Lanai or Maui.

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New locality records of Acarina and Mallophaga are added to the fauna of Antarctica. The following ectoparasites were received too late to include in the volume "Entomology of Antarctica" edited by Dr J. L. Gressitt (1967). All four species are reported in the above volume; however, not all are recorded from the following localities.

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Woodpeckers in Oregon. *Bird-Banding*, 48(2), 171-173.

Wilson, N., & Wodzicki, K. (1977). Ectoparasites from fruit bats and rats on Niue Island. *New Zealand Journal of Zoology*, 4(4), 383-387.

Examination of 3 species of mammals from Niue Island revealed the following host/ectoparasite associations. *Rattus rattus* (L.) : the mites *Androlaelaps casalis*, *Laelaps echidninus*, *L. nuttalli*, *Ornithonyssus bacoti*, *Listrophoroides cucullatus*, and *Dermatophagoides pteronyssinus*; and the louse *Polyplax spinulosa*. *Rattus exulans* (Peale) : *L. echidninus*, *L. nuttalli*, and *L. cucullatus*; and the louse *Hoplopleura pacifica*. *Pteropus t. tonganus* Quoy & Gaimard: the mite *Meristaspis calcarata*. These new records for the island are discussed in relation to existing records from the Pacific region.

Wilson, N. (1978). Four ectoparasites from South Dakota. *Entomological News*, 89.

Haas, G. E., R. E. Barrett, and N. Wilson. 1978. Siphonaptera from Mammals in Alaska. *Canadian Journal of Zoology* 56 (2): 333-338.

Wilson, N., & Oliver, G. V. (1978). Noteworthy records of two ectoparasites (Cimicidae and Hippoboscidae) from the turkey vulture in Texas. *The Southwestern Naturalist*, 23(2), 305-307.

Haas, G. E., T. Rumfelt, L. Johnson, and N. Wilson. 1979. Siphonaptera from Mammals in Alaska. Supplement I. *Canadian Journal of Zoology* 57 (9): 1822-1825.

Wilson, N., & Oliver Jr, G. V. (1979). New records of chewing lice (Mallophaga: Boopidae and Trichodectidae) from native mammals in Texas. *Southwestern Entomologist*, 4, 156-162.

Doster, G. L., Wilson, N., & Kellogg, F. E. (1980). Ectoparasites collected from bobwhite quail in the southeastern United States. *Journal of Wildlife Diseases*, 16(4), 515-520.

Wilson, N., & Haas, G. E. (1980). Ectoparasites (Mallophaga, Diptera, Acari) from Alaskan birds. 7. *Proceedings-Entomological Society of Washington*, 82.

Wilson, N. (1980). Records of ticks from Jamaica with the description of *Ixodes* (*Ixodes*) *copei* n. sp. from the mockingbird, *Mimus polyglottos* (Acari: Ixodidae). *International Journal of Acarology*, 6(2), 157-162.

A new species of tick, *Ixodes copei* is described from Jamaica. Hosts were the Mockingbird, *Mimus polyglottos* and White-eyed Thrush, *Turdus jamaicensis*. Records also are given for *Amblyomma cajennense* (Fabricius), *Anocentor nitens* (Neumann), *Boophilus microplus* (Canestrini), and *Rhipicephalus sanguineus* (Latreille).

Haas, G. E., & Wilson, N. (1981). BIRD FLEAS (SIPHONAPTERA: CERATOPHYLLIDAE) OF THE GENERA *DASYPSYLLUS* AND *MIOCTENOPSYLLA* IN ALASKA. *The Wasmann Journal of Biology*, 39, 72.

Haas, G. E. and N. Wilson. 1982. Fleas (Siphonaptera) from Nests of the Red Squirrel (*Tamiasciurus hudsonicus*) and Burrows of the Arctic Ground Squirrel (*Spermophilus parryii*) in Alaska. *Wasmann Journal of Biology* 40 (1-2): 59-65.

Lewis, R. E., & Wilson, N. (1982). A new species of *Nycteridopsylla* (Siphonaptera: Ischnopsyllidae) from southwestern United States, with a key to the North American species. *Journal of Medical Entomology*, 19(5), 605-614.

A new species of *NYCTERIDOPSYLLA* (Siphonaptera: Ischnopsyllidae) is described from the United States, and the other 2 North American taxa are discussed and illustrated. A summary of the biology, host preferences, and distribution of all 3 species is included, along with a key to both sexes.

Schwartz, C. C., R. Stephenson, and N. Wilson. 1983. *Trichodectes canis* on the Gray Wolf and Coyote on Kenai Peninsula, Alaska. *Journal of Wildlife Diseases* 19 (4): 372-373.

----. 1984. THE WASMANN JOURNAL OF BIOLOGY 42 (1-2), pp. 72-77.

Wilson, N. and R. L. Zarnke. 1985. Occurrence of the Ear Canker Mite, *Otodectes cynotis* (Hering), on the Wolverine, *Gulo gulo* (L.). *Journal of Wildlife Diseases* 21 (2): 180.

Wilson, N., G. C. Smith, and D. L. Sykes. 1986. An Additional Record of *Ornithocoris pallidus* (Hemiptera: Cimicidae) for the United States. *Journal of Medical Entomology* 23 (5): 575.

Haas, G. E., Wilson, N., & Brighton, H. D. (1986). Observations on Norway rats, *Rattus norvegicus*, in Kodiak, Alaska. *Canadian field-naturalist. Ottawa ON*, 100(3), 383-385.

Wilson, N., Dietz, J. M., & Whitaker, J. O. (1989). Ectoparasitic Acari found on golden lion tamarins (*Leontopithecus rosalia rosalia*) from Brazil. *Journal of wildlife diseases*, 25(3), 433-435.

Haas, G. E., N. Wilson, T. O. Osborne, R. L. Zarnke, L. Johnson, and J. O. Wolff. 1989. Mammal Fleas (Siphonaptera) of Alaska and Yukon Territory. *Canadian Journal of Zoology* 67 (2): 394-405.

Durden, L. A., Peterson, B. V., Wilson, N., & Christiansen, B. (1990). Some ectoparasites of bats from Seram Island, Indonesia. *Entomological News*, 101(1), 48-56.

New host and distribution records are provided for some ectoparasites of the families Streblidae (*Brachytarsina amboinensis*, *Megastrebla gigantea*, *M. parvior* and *Raymondia pseudopagodarum*), Nycteribiidae (*Archinycteribia actena* and *Nycteribia parilis*), Ischnopsyllidae (*Thaumapsylla breviceps*), Ixodidae (*Ixodes kopsteini* and *I. simplex*), Argasidae (*Ornithodoros* sp.), Laelapidae (*Neolaelaps spinosa*), Spinturnicidae (*Ancystropus taprobanius*, *A. zeleborii*, *Meristaspis calcarata*, *M. jordani*, *M. lateralis*,

Paraperiglischrus rhinolophus and *Spinturnix psi*) and Macronyssidae (*Macronyssus* sp. and *Trichonyssus* sp.) removed from 9 species of bats (*Dobsonia viridis*, *D. moluccensis*, *Rousettus amplexicaudatus*, *Syconycteris australis*, *Pteropus temmincki*, *Miniopterus australis*, *Rhinolophus euryotis*, *Chaerephon jobensis* and *Mormopterus beccarii*) collected in Mansuela National Park, on the island of Seram, Indonesia.

Durden, L. A., & Wilson, N. (1990). Ectoparasitic and phoretic arthropods of Virginia opossums (*Didelphis virginiana*) in central Tennessee. *The Journal of parasitology*, 581-583.

Thirteen species of ectoparasitic (12) or phoretic (1) arthropods were collected from 26 adult Virginia opossums, *Didelphis virginiana*, live-trapped from April through September 1987 in Davidson County, Tennessee. The cat flea *Ctenocephalides felis* and the American dog tick *Dermacentor variabilis* were the predominant species with respect to mean intensity and prevalence. The macronyssid mite *Ornithonyssus wernecki* and the atopomelid mite *Didelphilichus serratifer*, both specific parasites of this host, showed high intensities but low prevalences. Other fleas collected were *Cediopsylla simplex*, *Ctenophthalmus pseudagyrtes*, and *Orchopeas howardi*. The tick *Amblyomma americanum*, the myobiid mite *Archemyobia inexpectatus*, and the trombiculid (chigger) mites *Eutrombicula splendens*, *Leptotrombidium peromysci* (first record from this host) and *Neotrombicula cavicola* (first record from this host), were also recorded. One phoretic species, the glycyphagid mite *Marsupialichus brasiliensis*, was noted.

Wilson, N. A., S. R. Telford Jr., and D. J. Forrester. 1991. Ectoparasites of a Population of Urban Gray Squirrels in Northern Florida. *Journal of Medical Entomology* 28 (3): 461-464.

Durden, L. A., & Wilson, N. (1991). Parasitic and phoretic arthropods of sylvatic and commensal white-footed mice (*Peromyscus leucopus*) in central Tennessee, with notes on Lyme disease. *The Journal of parasitology*, 219-223.

Sixteen species of parasitic or phoretic arthropods were collected from 56 white-footed mice, *Peromyscus leucopus*, live-trapped in central Tennessee from April through November 1987. Arthropod infestation was compared for mice taken from sylvatic (woodland) versus commensal (household) habitats. Three species were recorded from hosts in both habitats: the sucking louse *Hoplopleura hesperomydis*, the flea *Epidemia wenmanni*, and the laelapid mite *Androlaelaps casalis*. Twelve of the 13 remaining arthropod species were taken only from mice trapped in woodland whereas the phoretic glycyphagid mite *Glycyphagus hypudaei* was collected only from commensal mice. Arthropod faunas on commensal hosts clearly were impoverished. The 12 additional arthropod species recorded from the woodland mice consisted of 1 nidicolous beetle, *Leptinus orientamericanus*; 1 bot, *Cuterebra fontinella*; 3 fleas, *Ctenophthalmus pseudagyrtes*, *Orchopeas leucopus* and *Peromyscopsylla scotti*; 1 tick, *Dermacentor variabilis*; 2 mesostigmatid mites, *Androlaelaps fahrenheitsi* and *Ornithonyssus bacoti*; 3 chiggers, *Comatacarus americanus*, *Euschoengastia peromysci*, and *Leptotrombidium peromysci*; and 1 undescribed pygmephorid mite of the genus *Pygmephorus*. Two

nymphal and 100 larval *D. variabilis* were examined for spirochetes and found to be uninfected.

McAllister, C. T., & Wilson, N. (1992). Two additions to the flea (Siphonaptera: Hystrichopsyllidae, Leptopsyllidae) fauna of Texas. *Texas Journal of Science*, 44(2), 245-247.

Peromyscopsylla scotti (n=2) and *Atyphloceras echis echis* (n=1) are reported from Texas, USA, for the first time, having being collected from a fulvous harvest mouse (*Reithrodontomys fulvescens laceyi*) and a southern plains woodrat (*Neotoma micropus micropus*), respectively, during January-February 1990.

DURDEN, L. A., BEST, T. L., WILSON, N., & HILTON, C. D. (1992). Ectoparasitic mites (Acari) of sympatric Brazilian free-tailed bats and big brown bats in Alabama. *Journal of medical entomology*, 29(3), 507-511.

Seven species of mites were recovered from 133 Brazilian free-tailed bats, *Tadarida brasiliensis*, and 94 big brown bats, *Eptesicus fuscus*, from February through November 1990 in colonies that shared roosting space in east-central Alabama. The macronyssid *Chiroptonyssus robustipes* (Ewing) was the most common mite on *T. brasiliensis* (964 mites, 87% of bats infested) and on *E. fuscus* (109 mites, 29% of bats infested). However, *C. robustipes* normally is a specific parasite of *T. brasiliensis*. The macronyssids *Steatonyssus ceratognathus* (Ewing) and *S. occidentalis* (Ewing) were recovered from both species of bats in low numbers. *S. ceratognathus* is not a typical parasite of either species of bat, but *S. occidentalis* normally is specific to *E. fuscus*. Predictably, *S. occidentalis* was most frequently collected from *E. fuscus* (16 mites, 9% of bats infested), but two specimens were recovered from *T. brasiliensis*. Five specimens of the laelapid *Androlaelaps casalis* (Berlese) (a mite that is frequently associated with rodents) and one specimen of the myobiid mite *Ewingana (Doreyana) longa* (Ewing) (a specific ectoparasite of *T. brasiliensis*) were also recovered from *T. brasiliensis*. Singletons of the rosensteiniids *Mydopholeus* sp. and *Nycteriglyphites pennsylvanicus* Fain, Lukoschus & Whitaker were the only additional mites collected from *E. fuscus*; both of these mites have previously been collected from bats or their guano but are recorded here from Alabama for the first time. With respect to ectoparasite cross-infestations, *E. fuscus* appears to be at greater risk from sharing roosts with *T. brasiliensis*. This is highlighted by the comparatively large numbers of *C. robustipes* that occurred on *E. fuscus* and the low numbers of *S. occidentalis* on *T. brasiliensis*. Although mites were the only arthropods recovered from bats in this study, a separate survey in 1991 revealed that the bat bug *Cimex adjunctus* Barber infested some other colonies of *T. brasiliensis* and *E. fuscus* in Alabama.

Haas, G.E. & N. Wilson. 1998. *Polygenis martinezbaezi* (Siphonaptera: Rhopalopsyllidae) Reared from a Rodent Nest found in the Peloncillo Mountains of Southwestern New Mexico. *Journal of Medical Entomology* 35 (4): 431-432.

Chilton, G., M. J. Vonhof, B. V. Peterson, and N. Wilson. 2000. Ectoparasitic Insects of

Bats in British Columbia, Canada. *Journal of Parasitology* 86 (1): 191-192.

One species of parasitic bug (Hemiptera: Cimicidae), 3 species of fleas (Siphonaptera: Ischnopsyllidae), and 2 species of parasitic flies (Diptera: Nycteribiidae) were collected from 9 species of bats (Chiroptera: Vespertilionidae) in southern interior and northeastern British Columbia, Canada. Female bats that return daily to maternity roosts were more frequently infested with both cimicids and ischnopsyllids than were male bats. Some differences in ectoparasite infestation can be attributed to differences in roosting behavior of the host. New national records for 2 parasite species, and 8 new host records are established for Canada.

Wilson, N. A., & Galloway, T. D. (2002). The Occurrence of the Bat Bug, *Cimex pilosellus* (Horváth)(Hemiptera: Cimicidae), in Manitoba, Canada. *Proceedings of the Entomological Society of Manitoba*, 58, 5.

Although bats are an ecologically important component of the mammalian fauna in much of Canada, their ectoparasites are seldom collected and poorly known. Many cimicids are blood-feeding ectoparasites of bats, and are found most frequently in the cracks and crevices in roosting areas of their hosts, rather than on the hosts themselves. Usinger's 1966 monograph of the cimicids is still, taxonomically and biogeographically, the most comprehensive treatment of the group. Apart from this monograph, most of the work on bat bugs is anecdotal and much of the information is widely scattered in the literature.

Wilson, N., & Durden, L. A. (2003). Ectoparasites of terrestrial vertebrates inhabiting the Georgia Barrier Islands, USA: an inventory and preliminary biogeographical analysis. *Journal of Biogeography*, 30(8), 1207-1220.

Aim: To document ectoparasitic arthropods of terrestrial vertebrates inhabiting barrier islands and to offer preliminary interpretations.

Location: The coast of Georgia, USA, with most data presented for St Catherines, Sapelo, Jekyll, and Cumberland islands.

Methods: Ectoparasites were collected and identified from hundreds of live-trapped mammals, birds, and reptiles. Ticks and louse-flies were also collected by dragging or flagging a white cloth through vegetation. Some ectoparasites were recovered from host nests or roosting sites. Literature records and specimen records from the US National Tick Collection supplemented these data.

Results: Inventories of ectoparasite species recovered from vertebrates on each island are provided. Many new records and a small number of undescribed species are reported.

Main conclusions: Compared with most ectoparasite faunas associated with the same host species on the adjacent mainland, ectoparasite species diversity on the barrier islands is depauperate. Possible reasons for this phenomenon are discussed including ecological, geological, and vicariant factors. Tick faunas do not appear to be depauperate on the islands probably because immature stages attach to visiting or migrating birds. Some

host-ectoparasite associations that we recorded are atypical for mainland faunas. This trend has also been recorded in some previous surveys of insular ectoparasites and may be related to vacant ectoparasite niches on some islands.

Haas, G. E., N. Wilson, and C. T. McAllister. 2004. Fleas (Siphonaptera: Ceratophyllidae, Ctenophthalmidae) from Rodents in Five Southwestern States. *Western North American Naturalist* 64 (4): 514-517.

McAllister, C. T., C. R. Bursey, and N. Wilson. 2006. Parasites of the Brazilian Free-Tailed Bat, *Tadarida brasiliensis* (Chiroptera: Molossidae), from Southwestern Arkansas. *Texas Journal of Science* 58 (1): 87-92.

Lewis, R. E., & Wilson, N. (2006). A Review of the Ceratophyllid subfamily Dactylopsyllinae. Part 2. *Dactylopsylla* Jordan, 1929, and *Foxella* Wagner, 1929 (Siphonaptera: Ceratophyllidae). *Annals of Carnegie Museum*, 75(4), 203-229.

There are three genera of fleas that are specific ectoparasites of the North American pocket gophers (Rodentia: Geomyidae): *Spicata* I. Fox, 1940, *Dactylopsylla* Jordan, 1929, and *Foxella* Wagner, 1929. All belong to the subfamily Dactylopsyllinae (Ceratophyllidae) and are restricted to the Nearctic Region of North and Central America. Seven species of *Dactylopsylla* Jordan, 1929, and the 14 named taxa of *Foxella* Wagner, 1929 are reviewed here with respect to their taxonomic affinities, distribution and host preferences. *Dactylopsylla bluei psilos* Prince and Stark, 1951, is a new synonym of *Dactylopsylla bluei* (C. Fox, 1909). *Foxella (Afoxella) macgregori* Barrera, 1953, is synonymized with *Foxella (Afoxella) mexicana* I. Fox, 1939. The remaining taxa are each briefly discussed with respect to their type data, nomenclatural history and non-genitalic characters. The following 11 names are new synonyms of *Foxella (Foxella) ignota* (Baker, 1895) and are not accorded subspecies status as done by previous authors: *Ceratophyllus franciscanus* Rothschild, 1910; *Ceratophyllus apachinus* C. Fox, 1914; *Foxella ignotus acutus* Stewart, 1940; *Ceratophyllus ignotus albertensis* Jordan and Rothschild, 1915; *Ceratophyllus ignotus reclusa* Jordan and Rothschild, 1915; *Foxella ignotus* [sic] *utahensis* Wagner, 1936; *Foxella ignota coufferi* Augustson, 1942; *Foxella ignota omissa* Prince, 1945; *Foxella utahensis arizonensis* Hubbard, 1947; *Foxella ignota clantoni* Hubbard, 1949; and *Foxella ignota chapmani* Hubbard, 1958. Keys to the genera, subgenera and species are presented and the entire assemblage is reviewed with respect to the distribution of the various taxa and their host preferences. The genus-group name *Foxelloides* Hubbard, 1943b, is noted as a synonym of *Spicata* I. Fox.

Hastriter, M. W., G. E. Haas, and N. Wilson. 2006. New Distribution Records for *Stenoponia americana* (Baker) and *Stenoponia ponera* Traub and Johnson (Siphonaptera: Ctenophthalmidae) with a Review of Records from the Southwestern United States. *Zootaxa* (1253): 51-59.

Whitaker, John O. Jr.; Walters, Brianne L. ; Castor, Linda K. ; Ritzi, Christopher M.; and Wilson, Nixon. 2007. "HOST AND DISTRIBUTION LISTS OF MITES (ACARI), PARASITIC AND PHORETIC, IN THE HAIR OR ON THE SKIN OF NORTH

AMERICAN WILD MAMMALS NORTH OF MEXICO: RECORDS SINCE 1974".
 Faculty Publications from the Harold W. Manter Laboratory of Parasitology. Paper 1.
<http://digitalcommons.unl.edu/parasitologyfacpubs/1>

Information concerning mites found in the hair of wild mammals of North America north of Mexico is summarized for records accumulated since a similar paper was published by Whitaker and Wilson in 1974. Chiggers, ticks and species that burrow into or under the skin are generally excluded although follicle-inhabiting mites are included. Included are lists a) of the mites organized taxonomically, b) hosts from which each species has been reported, and c) states and provinces citing references for each separate record.

Durden, L. A., Wilson, N., Eckerlin, R. P., & Baker, W. W. (2012). The flea (Siphonaptera) fauna of Georgia, USA: hosts, distribution and medical-veterinary importance. *Annals of Carnegie Museum*, 80(2), 83-113.

Host and distribution data are provided for the 26 species of fleas recorded from Georgia, U.S.A.: *Cediopsylla inaequalis*, *C. simplex* (rabbit flea), *Ctenocephalides canis* (dog flea), *C. felis* (cat flea), *Echidnophaga gallinacea* (sticktight flea), *Pulex irritans* (human flea), *P. simulans*, *Xenopsylla cheopis* (Oriental rat flea), *Polygenis gwyni*, *Conorhinopsylla stanfordi*, *Ctenophthalmus pseudadagyrtes*, *Doratopsylla blarinae*, *Epitedia cavernicola*, *E. wenmanni*, *Nearctopsylla georgiana*, *Stenoponia americana*, *Ceratophyllus celsus*, *Nosopsyllus fasciatus* (northern rat flea), *Orchopeas howardi* (squirrel flea), *O. leucopus*, *O. pennsylvanicus*, *Leptopsylla segnis* (European mouse flea), *Odontopsyllus multispinosus*, *Peromyscopsylla hesperomys*, *P. scotti* and *Sternopsylla distincta texana*. Three of these species, *P. gwyni*, *N. georgiana* and *O. howardi*, were described from Georgia specimens. While some of these flea species appear to be widely distributed throughout Georgia, *C. inaequalis* represents an introduction on hosts introduced for hunting from western North America, *D. blarinae*, *E. cavernicola*, *N. georgiana*, *C. celsus* and *P. hesperomys* were recorded only from Piedmont or mountainous regions in northern Georgia, and *P. gwyni* was mainly recorded in the Coastal Plain of southern Georgia. *Nearctopsylla georgiana* is known from only a single specimen and has never been recorded outside of northern Georgia. Fleas associated with domestic rats (*Rattus norvegicus* and *R. rattus*) were abundant in Georgia, especially coastal and southern Georgia during the 1930s–1950s but currently appear to be uncommon in the state. There were concerted control efforts against *Rattus* spp. and their fleas during that time period because of their reservoir and vector status, respectively, for *Rickettsia typhi*, the causative agent of murine (endemic) typhus. Human cases of murine typhus in Georgia have almost disappeared in recent decades. Other flea-borne diseases in Georgia include cat flea rickettsiosis caused by *Rickettsia felis*, bartonellosis (including cat scratch disease caused by *Bartonella henselae*), and sylvatic epidemic typhus caused by certain strains of *Rickettsia prowazekii*. Further, cat and dog fleas are intermediate hosts of the double-pored tapeworm, *Dipylidium caninum*, which typically infects dogs but can also infect cats and humans if infected fleas are inadvertently ingested. Human and pet-biting fleas such as the cat flea can also cause

flea-bite dermatitis in dogs, cats and humans. Poultry fleas such as *E. gallinacea* currently appear to be uncommon on domesticated birds in Georgia but this flea sometimes infests chickens, dogs and some other mammals in fairly large numbers.

McAllister, C. T., & Wilson, N. (2012). *Ctenophthalmus pseudagyrtes* (Siphonaptera: Ctenophthalmidae): New to the Flea Fauna of Texas. *The Southwestern Naturalist*, 57(3), 345-346.

Two female ctenophthalmid fleas, *Ctenophthalmus pseudagyrtes*, were collected in June 2003 from an eastern mole *Scalopus aquaticus* from Marion County, Texas. This flea originally was described from *S. aquaticus* in Michigan; *C. pseudagyrtes* has been reported from various mammalian hosts collected primarily east of the Mississippi River. This is the first report from Texas. In addition, we provide the second record of the mite *Haemogamasus harperi* in Texas from the same host.

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